

Unit 2

COMMUNICATION

64 items · 491 marks

Syllabus sections in this unit

2.1 Networks including the internet

Unit 2: Communication

1. M/J 26/P11/Q7/a,b,c

- 7 A networked computer game competition takes place in a school.

The game is hosted on a server on the school LAN (local area network). Each competitor logs on using their own computer.

- (a) The game runs as a thin-client.

Give **two** characteristics of a thin-client system.

1

2

[2]

- (b) One part of the school's LAN is a wired bus topology.

- (i) Explain how data is transmitted between two nodes in a bus topology using Ethernet.

[5]

- (ii) Give **two** drawbacks of using a bus topology.

1

2

[2]

- (c) Each of the school's computers has a static private IP address.

- (i) State what is meant by a static private IP address.

[1]

- (ii) The school's network uses subnetting.

Give **three** benefits of using subnetting in a network.

1

2

3

[3]

2. M/J 26/P12/Q6/d

- 6 A health monitor attaches to a person's arm and measures key health information, for example heart rate. The monitor transmits data to the user's smartphone. The results are analysed by the phone, which displays information to the user about their health.

- (d) The data from the health monitor is transmitted to the smartphone and stored in the cloud.

Discuss the benefits and drawbacks of storing the data from the health monitor in the cloud.

[4]

3. M/J 26/P13/Q3/c(ii,iii)

3 An animation is a sequence of bitmap images with a sound recording played at the same time.

(c) The sound and images are compressed independently before being transmitted over the internet to public cloud storage.

(ii) State what is meant by a public cloud.

[1]

(iii) Give **two** benefits to the animation creator of using public cloud storage for the animation.

1

2

[2]

4. O/N 25/P11/Q7/a,b,c

7 Web browsers use Internet Protocol (IP) addresses.

(a) Complete the table by describing the following types of IP address.

Type of IP address	Description
Static	
Public	

[2]

(b) Consider the following IP address:

256.0.0.A

Circle whether this IP address is IPv4, IPv6 or an invalid IP address.

IPv4

IPv6

Invalid

Justify your choice.

[2]

(c) A user types a Uniform Resource Locator (URL) into the address bar of a web browser to access a web page.

Explain how the web browser uses the URL to access the web page.

[4]

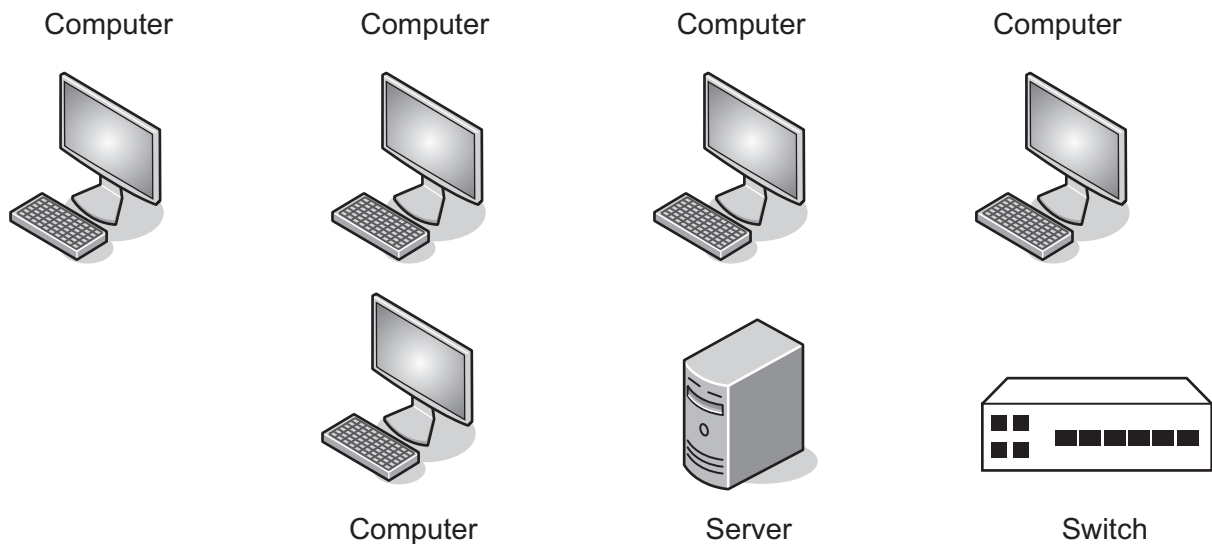
5. O/N 25/P12/Q5/a,b,c,d

5 A local area network (LAN) has five computers, one switch and one server.

(a) Describe the characteristics of a LAN.

[2]

(b) Complete the following diagram to show how these devices are connected in a star topology.



[2]

(c) Describe how packets are transmitted between two hosts using a star topology.

[3]

(d) Another type of network is a bus network.

Ethernet is used to transmit and receive data between the devices on the bus network.

Describe how collisions are detected and managed on this network.

[3]

6. O/N 25/P12/Q8/a

8 A restaurant accesses software and stores data using cloud computing.

(a) (i) Give **two** benefits of storing data using cloud computing.

1

2

[2]

(ii) Give **two** drawbacks of using cloud computing.

1

2

[2]

7. O/N 25/P13/Q4/a,b

- 4** A network engineer is setting up a Local Area Network (LAN) in an office. The network uses both wired and wireless connectivity.

(a) Give **three** advantages of a wireless network compared to a wired network.

- 1
- 2
- 3

[3]

(b) Give **one** drawback of using a wireless network.

[1]

8. M/J 25/P11/Q2/a,c

- 2** Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

(a) Explain how data is transferred using real-time bit streaming.

[4]

(c) The video conference is accessed over the internet.

Complete the table by stating how modems **and** dedicated lines are used when data is transmitted over the internet.

Hardware	Use
modems	
dedicated lines	

[2]

9. M/J 25/P12/Q6/a,b,c,d,e

6 A company has multiple sites in different cities.

The company has drivers who deliver products to customers. Each driver can connect to the company's WAN (Wide Area Network) whilst out of the office using a smartphone.

(a) State **two** ways that a WAN is different to a LAN (Local Area Network).

1

2

[2]

(b) The smartphones use the cell phone network to connect to the WAN.

Explain how data is transmitted using the cell phone network.

[4]

(c) Devices that connect to the WAN have an IP address.

Complete the following description of IP addresses by writing the missing words.

IPv4 is displayed as four groups of 8-bit numbers separated by

Each IPv4 address is 32 bits.

IPv6 is made of eight groups of 4 numbers separated by colons.

Multiple consecutive groups of can be replaced with a double colon.

Each IPv6 address is bits.

A IP address can change each time the computer connects to a network.

A IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.

[6]

(d) The main office for the company has four wired computers, one server, two printers, one central switch and a device providing a single point of access for the internet. The network is set up as a star topology.

Draw the topology for the main office network. Label all devices.

[4]

(e) Describe the role of a switch in a network.

[3]

10. M/J 25/P13/Q3/c

3 A programmer is writing a program in a high-level language.

(c) The computer program allows users to play a game in a virtual world. Users can play the game from any computer with internet access using a web browser. The game allows users to interact with other users.

Explain the reasons why the statement 'This computer game uses a client-server model' is correct.

[4]

11. O/N 24/P13/Q9/a,b,c,d

- 9** A computer is connected to a Local Area Network (LAN) that connects to a Wide Area Network (WAN).

(a) Describe the characteristics of a WAN.

[2]

(b) Copper cable can be used to transmit data in a network.

Complete the table by identifying **and** describing **two other** transmission media that can be used to transfer data in the WAN.

Transmission medium	Description

[4]

(c) The computer on the LAN is used for real-time video conferences, where people connected to the internet communicate in real-time using video and audio.

Explain how bit streaming is used in a real-time video conference.

[4]

(d) The LAN has a router. The router has a public IP address and a private IP address.

(i) State the purpose of a public IP address and a private IP address.

Public IP address

Private IP address

[2]

(ii) The LAN uses subnetting.

Describe the purpose of subnetting in a network.

[2]

12. M/J 24/P11/Q2/e

- 2** A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



- (e)** The video doorbell allows both real-time and on-demand bit streaming.

- (i)** State what is meant by bit streaming.

[1]

- (ii)** Give **two** differences between real-time and on-demand bit streaming.

1

2

[2]

13. M/J 24/P11/Q5/a

- 5** A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

- (a)** The system that allows customers to access their accounts using the application is a client-server model.

Describe the roles of the different devices in this model.

[4]

14. M/J 24/P11/Q8/a,b

8 A business is creating a local area network (LAN) in its office.

(a) The business is deciding which topology to use.

Tick (✓) **one or more** boxes in each row to identify the topology, or topologies, each statement describes.

Statement	Bus	Star	Mesh
all devices connect to one central device			
all devices connect to a central cable			
multiple paths for the packets to travel along			
robust against damage because if any line fails, the rest of the network retains full functionality			
most likely to lose data through collisions			

[5]

(b) The LAN will connect to the internet through a router. The router has a public IPv6 address.

(i) State why the router has a public IP address.

[1]

(ii) One difference between an IPv4 and IPv6 address is that the numbers in an IPv4 address are separated by full stops and in an IPv6 address they are separated by colons.

Identify **two other** differences between an IPv4 and IPv6 address.

1

2

[2]

15. M/J 24/P12/Q3/b,c

- 3 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.
- (b) The exam markers use software that operates as a thin-client to mark the exam papers.

Complete the table by identifying **two** characteristics of a thin-client.

Describe how each characteristic will be used in this software.

	Thin-client characteristic	Description of use in this software
1		
2		

[4]

- (c) Data transmitted on the internet passes through multiple different systems.
- (i) Describe the role of routers in the transmission of data through the internet.
- [2]
- (ii) Describe the role of the PSTN (Public Switched Telephone Network) in the transmission of data through the internet.

[2]

16. M/J 24/P13/Q5/a,b,c,d

- 5** A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

(a) The company has private cloud storage for its employees to store their work.

(i) Define the term **private cloud**.

[1]

(ii) Describe the benefits to the company of using private cloud storage instead of public cloud storage.

[3]

(b) Part of the internal structure of the wired LAN is a star topology.

Explain how packets are transmitted between two devices in a star topology.

[2]

(c) A different part of the network uses the Ethernet protocol.

(i) A collision is detected.

Describe how the collision is managed using Carrier Sense Multiple Access/Collision Detection (CSMA/CD).

[2]

(ii) Identify **two** drawbacks of using CSMA/CD.

1

2

[2]

(d) The devices in the office have static private IP addresses.

State what is meant by a **static private IP address**.

[1]

17. O/N 23/P11/Q2/a,b,c,d

2 A school has a Local Area Network (LAN).

(a) The LAN connects to the internet using a router.

Describe the function of a router in a network.

[3]

(b) Complete the following table by writing the purpose of each of these other hardware devices used to support the LAN.

Hardware device	Purpose
switch	
Wireless Access Point (WAP)	
bridge	

[3]

(c) The students can save their school files on a public cloud.

Identify **two** drawbacks of the students storing their files on the public cloud.

1

2

[2]

(d) A new classroom is being set up with 20 computers and a switch.

Explain **one** advantage of implementing a star topology instead of a bus topology in the new classroom.

[2]

18. O/N 23/P12/Q7/a,b,c,d

7 A Local Area Network (LAN) contains four devices:

- a router
- two laptop computers
- a server.

(a) The server has the IP address 192.168.3.2

Explain why this is **not** an IPv6 address.

[2]

(b) (i) The LAN is set up as a star topology.

Draw a diagram of the topology of the LAN.

[2]

(ii) Explain how data is transmitted between the two laptops in the LAN.

[2]

(iii) Subnetting can be used to separate a network into logical segments.

Describe **two other** reasons why subnetting is used in a network.

1

2

[4]

(c) State **three** tasks performed by devices to deal with collisions when using the Carrier Sense Multiple Access/Collision Detection (CSMA/CD) protocol in a network.

1

2

3

[3]

(d) The following incomplete table contains types of IP addresses and their descriptions.

Complete the table by writing the missing types of IP addresses and the missing descriptions.

Type of IP address	Description
	an IP address that is assigned to a device to allow direct access on the internet
static IP address	
	an IP address used for internal LAN communication only
dynamic IP address	

[4]

19. O/N 23/P13/Q3/a(i,ii)

- 3** A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (a) (i)** Explain why the company uses a public cloud to store these courses.

[2]

- (ii)** Describe **two** disadvantages of storing data on a public cloud compared to storing data on a server in a Local Area Network (LAN).

1

2

[4]

20. M/J 23/P11/Q4/d

- 4** A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (d)** The CCTV cameras are connected to a network and transfer their data wirelessly to the central computer.

- (i)** Each device on the network has an IP address.

Complete the description of IP addresses.

An IPv4 address contains groups of digits. Each group is represented in bits and the groups are separated by full stops.

An IPv6 address contains groups of digits. Each group is represented in bits. Multiple groups that only contain zeros can be replaced with a

[5]

- (ii)** The network makes use of subnetting.

Describe **two** benefits of subnetting a network.

1

2

[4]

21. M/J 23/P12/Q1/a,b,c,d,e

1 A company has a LAN (local area network).

(a) Give **two** benefits of connecting computers to a LAN.

1

2

[2]

(b) Give **two** characteristics of a LAN.

1

2

[2]

(c) One of the company's offices has one server and four computers connected in a star topology.

Draw a diagram to show the layout of the office's star topology.

[2]

(d) Computers can be connected using Ethernet.

Describe what is meant by **Ethernet**.

[3]

(e) The network runs as a thick-client model.

Describe what is meant by a **thick-client** model.

[2]

22. M/J 23/P13/Q2/a,b,c,d,e

- 2** A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).

(a) Identify **two** differences between a WAN and a LAN (local area network).

1

2

[2]

(b) The network uses different topologies in different areas of the sites. In one building there are five computers connected in a mesh topology.

(i) Describe what is meant by a mesh topology.

[2]

(ii) Give **two** advantages of using a mesh topology instead of a bus topology.

1

2

[2]

(c) The computers in one room are set up as thin-clients in a client-server model.

Describe the role of the different computers in this model.

[2]

(d) Students can connect their devices to the university network using cables or a wireless connection.

Explain the benefits to the students of allowing them to use **both** wired and wireless connections.

[4]

(e) One site has split the network into several subnetworks.

An IP address in a subnetwork is divided into two parts.

Identify **and** describe the **two** parts of an IP address in a subnetwork.

[3]

23. O/N 22/P11/Q8

- 8** A Local Area Network (LAN) uses a bus topology.

Describe how Carrier Sense Multiple Access/Collision Detection (CSMA/CD) is used in a bus network.

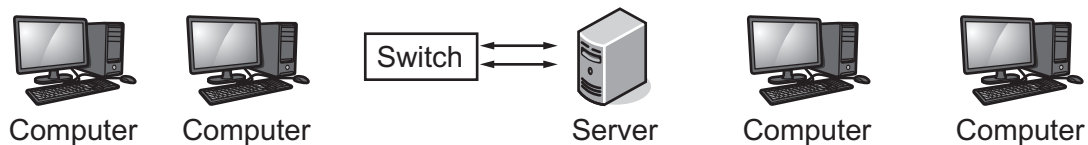
[4]

24. O/N 22/P12/Q10/a,b

10 A Local Area Network (LAN) consists of four computers, one server and a switch.

The LAN uses a star topology.

(a) Complete the following diagram to show how the hardware is connected.



[1]

(b) A router is attached to one of the devices on the LAN shown in **part (a)** to connect the LAN to the internet.

(i) Identify the device. Give a reason for your choice.

Device

Reason

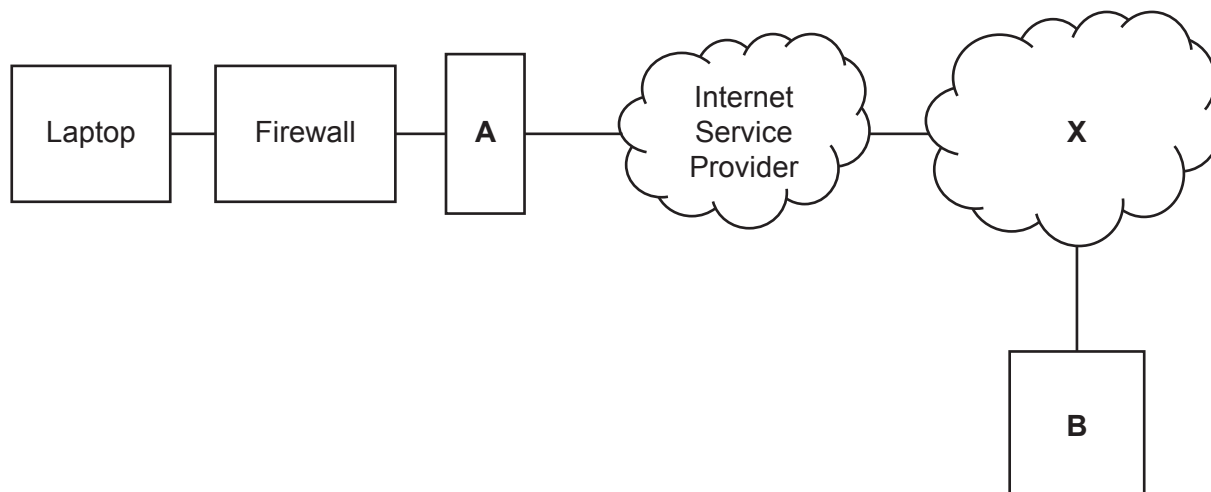
[2]

(ii) Describe the role **and** function of the router in the network.

[3]

25. O/N 22/P13/Q7/a,b,c

7 **(a)** The diagram shows the hardware and software used to retrieve files stored on the cloud.



Complete the following table by writing the answer for each row.

Answer

The name of device A that allows the laptop to connect to the internet	
A type of cloud, X	
An example of an application, B , that can run on the cloud	

[3]

- (b) Give **one** advantage and **two** disadvantages of transmitting data using satellites instead of copper cables.

Advantage 1

Disadvantage 1

Disadvantage 2

[3]

- (c) Local Area Networks (LANs) can be made up of several subnetworks.

- (i) Give **two** benefits of dividing a network into subnetworks by subnetting the LAN.

1

2

[2]

- (ii) A subnet mask is used when subnetting a LAN.

Two devices on the LAN are located in different subnetworks.

The IP addresses and corresponding subnet masks are shown:

Device IP address	Subnet mask
10.10.12.1	255.0.0.0
192.168.12.4	255.255.255.0

Identify the following network ID and host ID.

The network ID for the device with the IP address 10.10.12.1

The host ID for the device with the IP address 192.168.12.4

[2]

26. O/N 21/P11/Q8/a,b,c

8 A school is setting up a network within one of its buildings.

- (a) State whether the network will be a LAN (local area network) or a WAN (wide area network). Justify your choice.

[3]

- (b) One classroom in the building has 30 computers. The computers need to be connected to the network. Each computer has a network interface card (NIC).

Identify **two** possible devices that can be used to physically connect the 30 computers to the rest of the network.

1

2

[2]

- (c) The school has several laptops. Each laptop has a Wireless Network Interface Card (WNIC).

Describe the functions of a Wireless Network Interface Card.

[4]

27. O/N 21/P12/Q3/b

3 Andy likes to play computer games.

- (b) Andy plays some of the computer games over the internet. He has several devices that connect wirelessly to the router in his house.

- (i) Identify the topology of Andy's home network. Justify your choice.

Topology

Justification

[2]

- (ii) The router has a wireless access point (WAP) to allow the devices to connect wirelessly.

Identify **three** functions of the router in Andy's network.

1

2

3

[3]

28. O/N 21/P13/Q8/a,b,c

8 A school is setting up a network within one of its buildings.

- (a) State whether the network will be a LAN (local area network) or a WAN (wide area network). Justify your choice.

[3]

- (b) One classroom in the building has 30 computers. The computers need to be connected to the network. Each computer has a network interface card (NIC).

Identify **two** possible devices that can be used to physically connect the 30 computers to the rest of the network.

1

2

[2]

- (c) The school has several laptops. Each laptop has a Wireless Network Interface Card (WNIC).

Describe the functions of a Wireless Network Interface Card.

[4]

29. 9608 O/N 21/P11/Q11/a,b

11 The Internet uses a client-server model.

- (a) Describe the role of clients and servers on the Internet.

[2]

- (b) Computers on the Internet have IP addresses.

Describe the format of an IP address.

[3]

30. M/J 21/P11/Q4/a,b,c,d

4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

(a) Describe the key features of a peer-to-peer network.

[2]

(b) Describe **two** drawbacks to Melinda and her friends of using a peer-to-peer network.

1

2

[4]

(c) Melinda connects her laptop to the internet through her router.

(i) Tick (✓) **one** box in each row to identify whether the task is performed by the router or not.

Task	Performed by router	Not performed by router
Receives packets from devices		
Finds the IP address of a Uniform Resource Locator (URL)		
Directs each packet to all devices attached to it		
Stores the IP and/or MAC address of all devices attached to it		

[2]

(ii) Melinda mainly uses the internet to watch films and play computer games.

Tick (✓) **one** box to identify whether Melinda should connect to the router using a wired or wireless network **and** justify your choice.

Wired	
Wireless	

Justification

[3]

(d) Melinda sends emails from her webmail account (email account accessed through a website).

Explain whether Melinda is using the internet, or the World Wide Web (WWW), or both.

[3]

31. M/J 21/P12/Q5/c,d

5 Seth uses a computer for work.

(c) Seth accesses both software and data using cloud computing.

(i) Give **two** benefits of storing data using cloud computing.

1

2

[2]

(ii) Give **two** drawbacks of Seth using cloud computing.

1

2

[2]

(d) Draw **one** line from each term to its **most appropriate** description.

Term

Description

Public IP address

It is only visible to devices within the Local Area Network (LAN)

Private IP address

It increments by 1 each time the device connects to the internet

Dynamic IP address

A new one is reallocated each time a device connects to the internet

Static IP address

It can only be allocated to a router

It is visible to any device on the internet

It does not change each time a device connects to the internet

[4]

32. M/J 21/P13/Q4/a,b,c,d

4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

(a) Describe the key features of a peer-to-peer network.

[2]

(b) Describe **two** drawbacks to Melinda and her friends of using a peer-to-peer network.

1

2

[4]

(c) Melinda connects her laptop to the internet through her router.

(i) Tick (✓) **one** box in each row to identify whether the task is performed by the router or not.

Task	Performed by router	Not performed by router
Receives packets from devices		
Finds the IP address of a Uniform Resource Locator (URL)		
Directs each packet to all devices attached to it		
Stores the IP and/or MAC address of all devices attached to it		

[2]

(ii) Melinda mainly uses the internet to watch films and play computer games.

Tick (✓) **one** box to identify whether Melinda should connect to the router using a wired or wireless network **and** justify your choice.

Wired	
Wireless	

Justification

[3]

(d) Melinda sends emails from her webmail account (email account accessed through a website).

Explain whether Melinda is using the internet, or the World Wide Web (WWW), or both.

[3]

33. 9608 M/J 21/P11/Q9/a,b

- 9 (a) Identify **two** differences between a public IP address and a private IP address.

1

2

[2]

- (b) Complete the table by identifying the **most appropriate** term for each description. Each term must be different.

Description	Term
Receives data packets from a network and forwards them onto a similar network	
Manages access to a centralised resource	
Joins networks that use different sets of rules to transmit data	
Monitors and controls incoming and outgoing network traffic based on set criteria	

[4]

34. 9608 M/J 21/P12/Q9/a,b

- 9 Networks can be either wired using cables or wireless using radio waves.

- (a) Describe **one** benefit of using a wireless network compared to using a wired network.

[2]

- (b) Describe **two** drawbacks of using a wireless network compared to using a wired network.

Drawback 1

Drawback 2

[4]

35. 9608 M/J 21/P13/Q6/b

- (b) The web page 9608.html is accessed from the URL:

`https://www.cambridgeinternational.org/9608.html`

An employee of the company states, "A Domain Name and an IP address are exactly the same thing".

State whether this statement is true or false **and** justify your choice.

[3]

36. M/J 20/P12/Q7/a,b

- 7** A company allows customers to stream music from its servers over the Internet. The company's internet connection is currently provided through copper cables.

(a) Identify **two** pieces of hardware, other than the cables, that enable the servers to connect to the Internet. Describe the purpose of each device.

Device 1

Purpose

Device 2

Purpose

[4]

(b) The company wants to upgrade their internet connection to fibre-optic cables.

Give **one** benefit and **one** drawback to the company of upgrading to fibre-optic cables.

Benefit

Drawback

[2]

37. M/J 20/P13/Q1/a,b,c

- 1 Ana owns a small company with four employees. The office has a network containing several computers that run on a client-server model. There is one server that connects to the Internet using a router.

(a) Networks transmit data using various types of connection shown in the following table.

Complete the table.

Type of connection	Description
Fibre-optic	
	A communication device in Earth's orbit that receives and transmits data
Radio waves	
	Carries data as electrical signals and can consist of a twisted pair

[4]

- (b) Explain how the client-server model enables the employees to access the same files from different computers.

[2]

- (c) Each computer in the network has a private IP address.

Give **two** reasons why the computers do **not** have public IP addresses.

1

2

[2]

38. O/N 19/P11/Q4/a,c

4 Customers of a bank can access their account information by logging in on the bank's website.

(a) The bank has a client-server model of networked computers.

(i) Describe, using the bank as an example, the key features of a client-server model.

[3]

(ii) Give **two** other examples of applications that can use the client-server model.

1

2

[2]

(c) The bank is upgrading its local area network (LAN) copper cables to fibre-optic cables.

(i) State **two** benefits to the bank of upgrading to fibre-optic cable from copper cable.

1

2

[2]

(ii) State **two** drawbacks of upgrading to fibre-optic cables.

1

2

[2]

39. O/N 19/P12/Q6/e(i,ii)

6 Dominic uses a tablet computer to complete work. He records videos of his work to send to his colleagues to watch at a later date.

(e) Dominic sends his videos to his colleagues over the Internet using bit streaming.

(i) Describe how the video is sent using bit streaming.

[4]

(ii) Circle either Real-time or on-demand to identify whether the video will be sent using real-time or on-demand bit streaming. Justify your choice.

Real-time / on-demand

Justification

[2]

40. O/N 19/P13/Q1/c(i)

- 1 In a supermarket, a self-checkout machine allows customers to scan the barcodes of products and then pay for their shopping. These are an alternative to the traditional cashier-staffed checkout.



- (c) The self-checkout machines connect to a server that stores all the data for the supermarket. This is a client-server network.
- (i) Describe, using an example for the supermarket, the client-server network model.

[4]

41. M/J 19/P11/Q1/a,b,c

- 1** Devices connected to the Internet have IP (Internet Protocol) addresses.

- (a)** Three IPv4 addresses are given.

Circle either Valid or Invalid to indicate whether each address is valid or invalid. Explain your decision.

Address 1: 3A.21.2H.1 Valid / Invalid

Explanation

Address 2: 299.53.2.2 Valid / Invalid

Explanation

Address 3: 192.2.1.0 Valid / Invalid

Explanation

[3]

- (b)** A website can be accessed using either the Uniform Resource Locator (URL) or the IP address.

Describe how a URL is converted into its matching IP address.

[3]

- (c)** People use the Internet to stream media.

Complete the following statements by filling in the names of the missing methods of bit streaming.

..... bit streaming is used when watching a live stream of events that are currently taking place. The event is captured live with a video camera connected to a computer, and it cannot be paused or rewound.

..... bit streaming is used when watching an event that has taken place in the past. Existing media are encoded to bit streaming format and uploaded to a server. It can be paused and rewound.

[2]

42. M/J 19/P12/Q1/a,b,d

1 Computers on the Internet have IP addresses.

(a) IP addresses can be in either IPv4 or IPv6 format.

(i) Give an example of a valid IPv4 address.

[1]

(ii) State why there is a need for IPv6 addressing.

[1]

(iii) A computer's IPv6 address is:

C100:2235::1000:25AA:AA50

Explain why this IPv6 address would be an invalid IPv4 address.

[2]

(b) A company has computers in two separate buildings that communicate using the Internet over a Public Switched Telephone Network (PSTN).

(i) Describe the transmission of data using a PSTN.

[2]

(ii) The company wants to install a dedicated line between the two buildings.

Identify **one** benefit and **one** drawback of installing a dedicated line between the two buildings.

Benefit

Drawback

[2]

(d) The company has an email server.

Identify **three** other types of server.

1

2

3

[3]

43. O/N 18/P11/Q2/a,b,c

2 Computer **A** needs to access a web page.

- (a)** State how Computer **A** could access the web page without using a Domain Name Service (DNS).

[1]

- (b) (i)** The following table shows four IPv6 addresses.

State if each address is valid or invalid.

IP address	Valid or invalid
21E5:69AA:FFFF:1:E100:B691:1285:F56E	
::255.255.255.255	
59FB::1005:CC57:6571	
56FE::2159:5BBC::6594	

[4]

- (ii)** The following table shows four statements about either public or private IP addresses.

Tick (✓) **one** box in each row to indicate whether each statement refers to a public or a private IP address.

Statement	Public	Private
192.168.2.1 is an example of this type of address		
Assigned by the Internet Service Provider (ISP)		
IP address cannot be duplicated in different networks		
Network Address Translation (NAT) is necessary to access the Internet directly		

[4]

- (c)** One type of transmission media is copper cable.

Give **two** additional types of transmission media.

1

2

[2]

44. O/N 18/P12/Q2/a,b,c

2 Gopal types the Uniform Resource Locator (URL) of a website into a web browser.

(a) The following sequence (1 to 5) describes the steps that take place. There are three missing statements.

1 Gopal types into the web browser.

2

3 DNS looks up the URL in table

4

5

Three statements **A**, **B** and **C** are used to complete the sequence.

A	DNS finds corresponding IP address
B	Web browser sends URL to Domain Name Service (DNS)
C	DNS returns IP address to web browser

Write one of the letters **A** to **C** in the appropriate rows (2, 4 and 5) to complete the sequence. [2]

(b) Describe the purpose of an IP address.

[2]

(c) A telecommunications operator has installed fibre-optic cables in Gopal's neighbourhood.

(i) Give **three** benefits of fibre-optic cable over copper cable.

1

2

3

[3]

(ii) Give **two** drawbacks of fibre-optic cable over copper cable.

1

2

[2]

45. O/N 18/P13/Q4/a,b,c

4 Ava needs to view a website and she knows the Uniform Resource Locator (URL).

(a) Complete the series of steps that take place.

Write the **letter** of the appropriate statement in each space.

A	DNS finds corresponding IP
B	DNS looks up URL in table
C	Ava types the URL into a web browser

1

2 Web browser sends URL to Domain Name Service (DNS)

3

4

5 DNS returns IP address to web browser

[2]

(b) (i) An IPv4 address has been entered as 12.258.3

Give **two** reasons why this IP address is invalid.

1

2

[2]

(ii) An IPv6 address has been entered as 15EF:5L63::2014:BB::60AA

Give **two** reasons why this IP address is invalid.

1

2

[2]

(c) The table shows four descriptions of IP addresses.

Tick (✓) **one** box in each row to identify whether each description applies to a public or private IP address.

Description	Public	Private
The address can be reached over the Internet.		
The address is more secure.		
The address can only be accessed through the same LAN.		
The address can be duplicated in different networks.		

[4]

46. O/N 18/P13/Q7/a,b

- 7 The network manager of a Local Area Network (LAN) has replaced the Ethernet cables with a wireless network.

(a) Give **three** benefits of a wireless network compared to a wired network.

1

2

3

[3]

(b) Give **one** drawback of a wireless network compared to a wired network.

[1]

47. M/J 18/P11/Q1

- 1 Four communication media and five features are shown.

Draw one or more lines from each communication media to the appropriate feature(s).

Communication media

Fibre-optic cable

Radio waves

Copper cable

Satellite

Feature

Can be twisted pair or co-axial

Transmits light pulses

Large range of wavelengths

Least likely to have interference

Wireless transmission

[6]

48. M/J 18/P11/Q5/a,b

5 A college has a client-server network.

(a) The college has a file server and other servers.

State the purpose of **two** other servers in the college network.

Server 1

Server 2

[2]

(b) The students use the network to access the Internet.

One student stated, '*The Internet and the World Wide Web are the same thing*'.

Tick (✓) **one** box to indicate whether this statement is true or false.

True	False

Justify your choice.

[5]

49. M/J 17/P12/Q6/a,b

6 Downloading a file from a website is an example of a client-server application.

(a) Describe what is meant by the term **client-server** for this application.

[2]

(b) The following sequence of steps (1 to 5) describes what happens when someone uses their personal computer (PC) to request a web page. The web page consists of HTML tags and text content only. Four of the statements from **A**, **B**, **C**, **D**, **E** and **F** are used to complete the sequence.

A	Browser software interprets the script, renders the page and displays.
B	Browser software renders the page and displays.
C	Browser software compiles the script, renders the page and displays.
D	The web server retrieves the page.
E	The Domain Name Service (DNS) uses the domain name from the browser to look up the IP address of the web server.
F	The web server sends the web page content to the browser.

Write one of the letters A to F in the appropriate row to complete the sequence.

1. The user keys in the Uniform Resource Locator (URL) into the browser software.
- 2.
- 3.
- 4.
- 5.

[4]

50. O/N 16/P11/Q6

6 A user watches a video available on a website. The website uses on-demand bit streaming.

Describe how it is possible to watch the video without it continually pausing.

[4]

51. O/N 16/P13/Q6

6 A user watches a video available on a website. The website uses on-demand bit streaming.

Describe how it is possible to watch the video without it continually pausing.

[4]

52. M/J 16/P11/Q7/a,b,c

7 Access to World Wide Web content uses IP addressing.

(a) State what IP stands for.

[1]

(b) The following table shows four possible IP addresses.

Indicate for each IP address whether it is valid or invalid and give a reason.

Address	Denary / Hexadecimal	Valid or Invalid	Reason
3.2A.6AA.BBBB	Hexadecimal		
2.0.255.1	Denary		
6.0.257.6	Denary		
A.78.F4.J8	Hexadecimal		

[4]

(c) Describe **two** differences between public and private IP addresses.

1

2

[2]

53. M/J 16/P12/Q7/a,b,c

7 Access to World Wide Web content uses IP addressing.

(a) State what IP stands for.

[1]

(b) The following table shows four possible IP addresses.

Indicate for each IP address whether it is valid or invalid and give a reason.

Address	Denary / Hexadecimal	Valid or Invalid	Reason
3.2A.6AA.BBBB	Hexadecimal		
2.0.255.1	Denary		
6.0.257.6	Denary		
A.78.F4.J8	Hexadecimal		

[4]

(c) Describe **two** differences between public and private IP addresses.

1

2

[2]

54. M/J 16/P13/Q6/a,b,c,d,e

- 6 (a)** Explain the difference between the World Wide Web (WWW) and the Internet.

[2]

- (b)** Three methods of connecting devices include fibre-optic cables, copper cables and radio waves. The table below gives descriptions relating to these connection methods.

Tick (✓) one box on each row to show the method that best fits each description.

Description	Fibre-optic cable	Copper cable	Radio waves
Wireless medium			
Twisted-pair is an example			
Uses light waves			
WiFi			
Fastest transmission medium			

[5]

- (c)** Bit streaming is used for both real-time and on-demand services.

Describe **one** difference between real-time and on-demand bit streaming.

[2]

- (d)** A device needs an IP address to connect to the Internet. IPv4 is the more common type of IP address.

Describe, using an example, the format of an IPv4 address.

[3]

- (e)** A computer user keys in the Uniform Resource Locator (URL) of a web page into a web browser.

Describe how the browser uses the Domain Name Service (DNS) to display the web page.

[4]

55. O/N 15/P11/Q1/a,b,c

- 1 (a) Explain the term bit streaming.

[2]

- (b) A person watches a film streamed from a website on a tablet computer.

- (i) Give **two** benefits of using bit streaming for this purpose.

1

2

[2]

- (ii) State **two** potential problems of using bit streaming for this purpose.

1

2

[2]

- (c) Explain the terms on-demand bit streaming and real-time bit streaming.

[4]

56. O/N 15/P11/Q3/a,b(i)

- 3 (a) The table shows four statements about IP addresses.

Tick (✓) to show which of the statements are true.

Statement	True (✓)
The IP address consists of any number of digits separated by single dots (.)	
Each number in an IP address can range from 0 to 255	
IP addresses are used to ensure that messages and data reach their correct destinations	
Public IP addresses are considered to be more secure than private IP addresses	

[2]

- (b) Consider the URL:

<http://cie.org.uk/computerscience.html>

- (i) Give the meaning of the following parts of the URL.

http

cie.org.uk

computerscience.html

[3]

57. O/N 15/P11/Q6

- 6** A company operates a chemical plant, which has a number of processes. Local computers monitor these processes and collect data.

The computers transfer these data to a central computer 50km away. A telecommunications company (telco) provides cables.

Engineers at the telco had to decide which type of cable to use. They considered the use of either copper cable or fibre optic cable.

State **two** benefits of each type of cable. Each benefit must be clearly different.

Benefits of copper cable

1

2

Benefits of fibre optic cable

1

2

[4]

58. O/N 15/P11/Q7/a,b

7 (a) (i) Describe what is meant by a client-server model of networked computers.

[2]

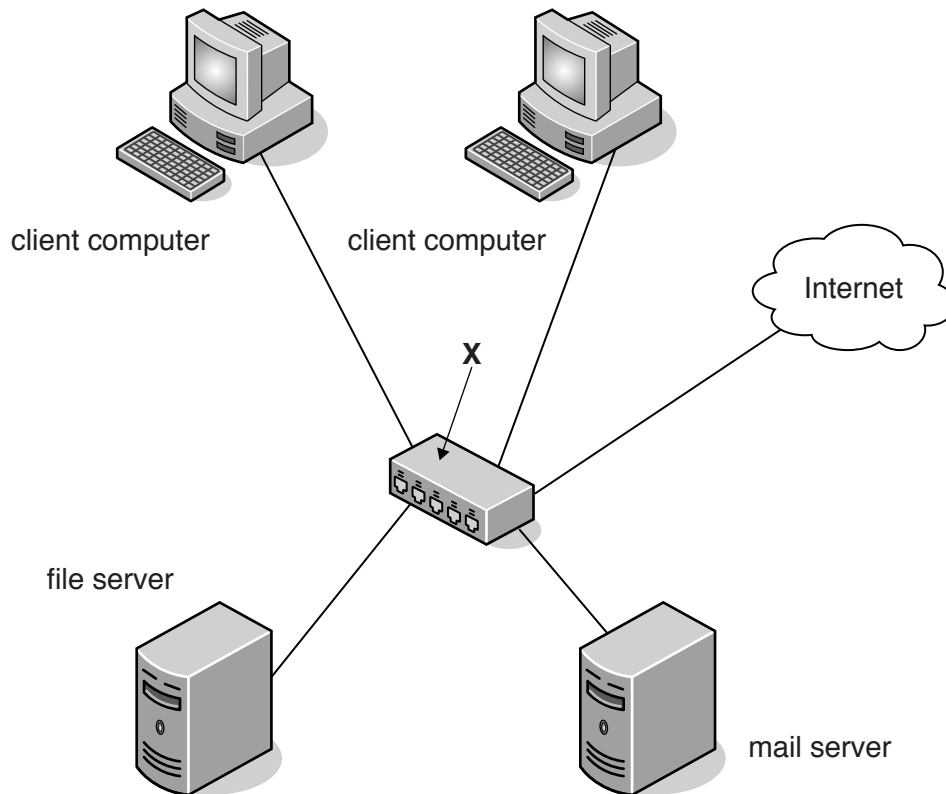
(ii) Give **two** benefits of using the client-server model.

1

2

[2]

(b) The diagram shows a computer network with connection to the Internet.



Name the hardware device labelled **X**.

[1]

59. O/N 15/P13/Q1/a,b,c

- 1 (a) Explain the term bit streaming.

[2]

- (b) A person watches a film streamed from a website on a tablet computer.

- (i) Give **two** benefits of using bit streaming for this purpose.

1

2

[2]

- (ii) State **two** potential problems of using bit streaming for this purpose.

1

2

[2]

- (c) Explain the terms on-demand bit streaming and real-time bit streaming.

[4]

60. O/N 15/P13/Q3/a,b(i)

- 3 (a) The table shows four statements about IP addresses.

Tick (✓) to show which of the statements are true.

Statement	True (✓)
The IP address consists of any number of digits separated by single dots (.)	
Each number in an IP address can range from 0 to 255	
IP addresses are used to ensure that messages and data reach their correct destinations	
Public IP addresses are considered to be more secure than private IP addresses	

[2]

- (b) Consider the URL:

<http://cie.org.uk/computerscience.html>

- (i) Give the meaning of the following parts of the URL.

http

cie.org.uk

computerscience.html

[3]

61. O/N 15/P13/Q6

- 6** A company operates a chemical plant, which has a number of processes. Local computers monitor these processes and collect data.

The computers transfer these data to a central computer 50km away. A telecommunications company (telco) provides cables.

Engineers at the telco had to decide which type of cable to use. They considered the use of either copper cable or fibre optic cable.

State **two** benefits of each type of cable. Each benefit must be clearly different.

Benefits of copper cable

1

2

Benefits of fibre optic cable

1

2

[4]

62. O/N 15/P13/Q7/a,b

7 (a) (i) Describe what is meant by a client-server model of networked computers.

[2]

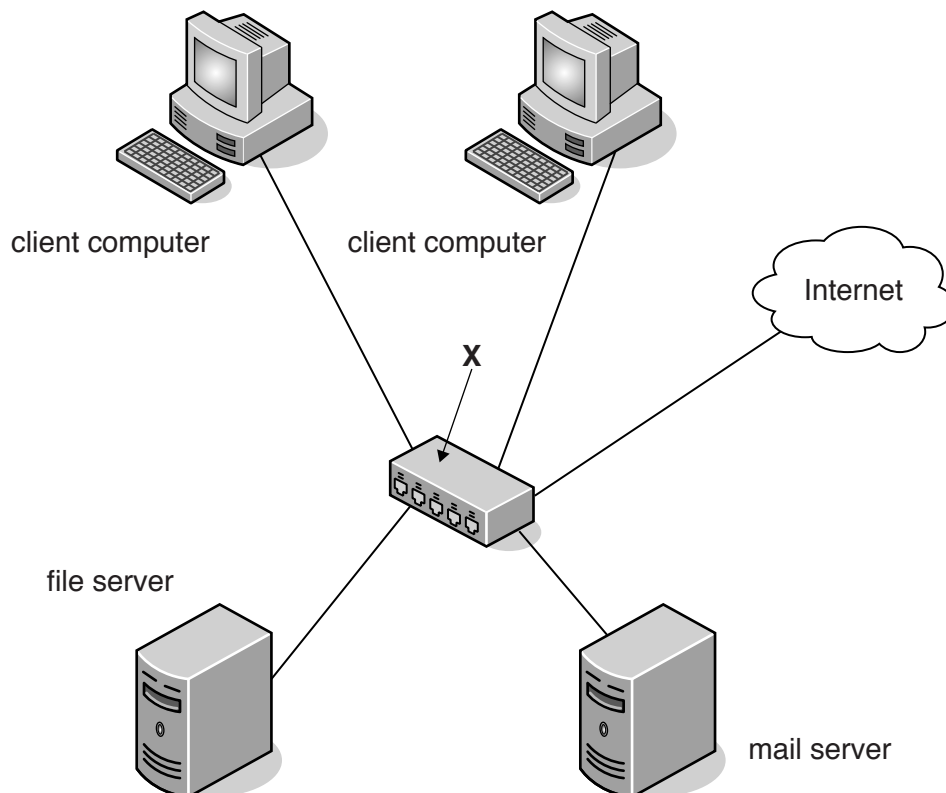
(ii) Give **two** benefits of using the client-server model.

1

2

[2]

(b) The diagram shows a computer network with connection to the Internet.



Name the hardware device labelled **X**.

[1]

63. M/J 15/P11/Q5/b,c(i,iii)

(b) Distinguish between the Internet and the World Wide Web (WWW).

[3]

(c) Name the hardware device that is being described:

(i) A device that transfers data from one network to another in an intelligent way. It has the task of forwarding data packets to their destination by the most efficient route.

[1]

(iii) A device or software that provides a specific function for computers using a network. The most common examples handle printing, file storage and the delivery of web pages.

[1]

64. M/J 15/P12/Q5/b,c(i,iii)

(b) Distinguish between the Internet and the World Wide Web (WWW).

[3]

(c) Name the hardware device that is being described:

(i) A device that transfers data from one network to another in an intelligent way. It has the task of forwarding data packets to their destination by the most efficient route.

[1]

(iii) A device or software that provides a specific function for computers using a network. The most common examples handle printing, file storage and the delivery of web pages.

[1]

Answers Section

1. M/J 26/P11/Q7/a,b,c

7(a)	1 mark for each point to max 2 - Data is not stored on the client computer // all data is stored on the server - The client computer is reliant on access to server - The client computer is heavily reliant on the network connection - The client computer requires few local resources, for example, memory / processing power / storage - The client computer performs minimal functions / processes	2	
7(b)(i)	1 mark for each point to max 5 Bus: - Both nodes are connected to a bus/cable - Data is broadcast on the cable - but is only accepted by the intended recipient Ethernet: - Uses the IP/MAC addresses of the two nodes - Ethernet transmits data as frames - Ethernet uses Carrier Sense Multi-Access / Collision Detection / CSMA/CD - The transmitting node listens to see if the line is free and only submits data when the line is free - The node waits a random time if a collision is detected and then tries to send again	5	Award a maximum of 4 marks if only Ethernet referenced
7(b)(ii)	1 mark for each point to max 2 - Bus networks are inefficient with a large number of nodes - because only one transmission can be on the line at once - Bus networks are inefficient with large numbers of simultaneous data transmissions - because of the need for nodes to wait for the line to be free - Bus networks are not easily scalable - If the bus/cable goes down the whole/part of the network is unusable - Transmissions are broadcast to all nodes, so could be privacy/confidentiality/security issues	2	The answer spaces are numbered, so mark the first answer in each space
7(c)(i)	1 mark for a complete answer It does not change and it is only visible/accessible within the network/LAN	1	The answer needs both static and private
7(c)(ii)	1 mark for each point to max 3 For example: - Subnetting splits a larger network into smaller/more manageable parts - The network has increased efficiency – just faster is NE - Congestion is reduced // fewer collisions happen - Security is improved - It can be more straightforward to isolate faults - Easier to scale a network with subnets	3	Accept any sensible benefit.

2. M/J 26/P12/Q6/d

6(d)	1 mark for each correct point to max 4, max 3 if only either benefits or drawbacks discussed, not both For example: - The data can be accessed anywhere with internet access // without internet access the data cannot be accessed - The health monitor and smart phone can be mobile and still upload data - The data is shareable with medical professionals - Storing data in the cloud reduces the need for large amounts of internal storage on the phone - The user can upgrade/change their device and retain old data // the data can be accessed from different appropriate devices - The user has no control over security // There could be security issues with storing the data on the cloud - The data could be sold or misused by the cloud owner	4	Note: The mark points are examples only. Accept any sensible benefit or drawback Any answer about cost must be suitably qualified. Answers about no control need to be qualified Do not award two marks for a benefit and it's reverse as a drawback Just easier for hackers is NE
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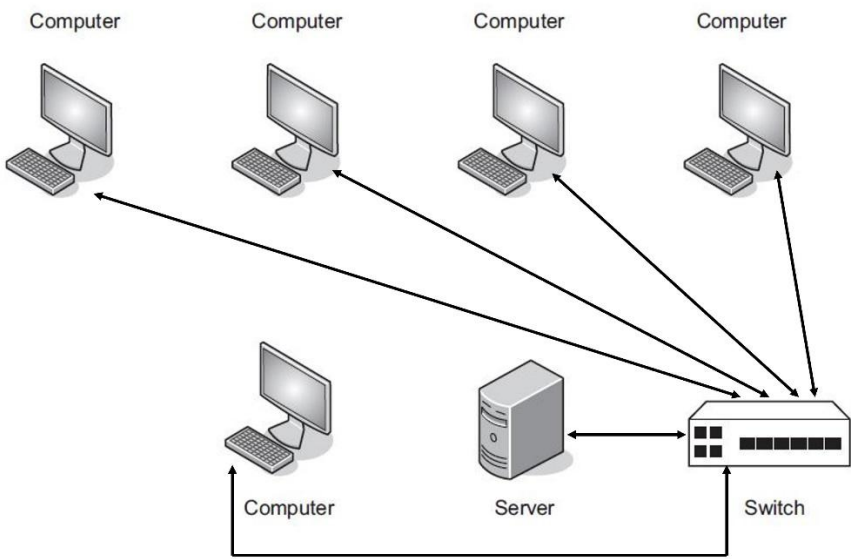
3. M/J 26/P13/Q3/c(ii,iii)

3(c)(ii)	1 mark for: Storage located on a remote server that anyone can sign up to and use	1	Not commercial sites
3(c)(iii)	1 mark each to max 2 - Animator can access anywhere that has internet access // animator can work on the animation in different locations as long as they have internet - Animator can access from any device // animator can work on different devices - Animator can give authentication/access to selected people without having to transfer the files/animation individually // Online collaboration with colleagues - Storing data in the cloud reduces the need for large amounts of internal storage on the local devices - The cloud company manages backups etc. - The Animator not responsible for security - More storage capacity readily available if needed	2	

4. O/N 25/P11/Q7/a,b,c

7(a)	1 mark for each correct description, max 2 marks	2						
	<table><tr><th>Type of IP address</th><th>Description</th></tr><tr><td>Static</td><td>IP address that does not change each time a device connects to the network</td></tr><tr><td>Public</td><td>IP address assigned to a device on a network to allow the device to be visible on the internet</td></tr></table>	Type of IP address	Description	Static	IP address that does not change each time a device connects to the network	Public	IP address assigned to a device on a network to allow the device to be visible on the internet	
Type of IP address	Description							
Static	IP address that does not change each time a device connects to the network							
Public	IP address assigned to a device on a network to allow the device to be visible on the internet							
7(b)	1 mark per bullet point for justification, max 2 marks Choice: Invalid (no mark for the choice) Justification: - It is not IPv4 because the maximum denary value should be 255, this uses 256 - It is not IPv6 because it uses a full stop to separate sections and not colon // It is not IPv6 because there are not enough groups	2						
7(c)	1 mark per bullet point, max 4 marks - The browser checks its cache for the URL - The browser parses the URL and splits it into its component parts - The browser finds the IP address for the domain name by querying a DNS server - The browser receives the matching IP address from the DNS server - The browser creates connection with the web server with the matching IP address - A request for the resource is sent to the web server with that IP address - The browser renders and displays the result - The IP address is stored in the browser cache for future use	4						

5. O/N 25/P12/Q5/a,b,c,d

5(a)	1 mark per bullet point, max 2 marks - Covers a small geographical area - Normally privately owned / dedicated infrastructure - Can use wired or wireless communication	2
5(b)	1 mark per bullet point, max 2 marks - All computers directly connected to switch and no other connections - Switch connected directly to server  The diagram illustrates a star network topology. Four desktop computers are arranged in a horizontal row at the top. Below them, in the center, is a server tower. To the right of the server is a network switch. Arrows point from each of the four computers to the switch. A double-headed arrow connects the server and the switch, indicating bidirectional communication.	2
5(c)	1 mark per bullet point, max 3 marks - The sending device sends the packets to the switch - The switch checks the destination address on each packet ... and sends them only to the intended receiver	3
5(d)	1 mark per bullet point, max 3 marks - The use of <u>CSMA/CD</u> // Carrier Sense Multiple Access with Collision Detection - The workstations listen to the communication channel ... and send data only when there is no data being transmitted / the line is idle - If a collision is detected, transmission is aborted ...and a jamming signal is sent - The workstation calculates a random wait time before trying to re-transmit - The random time is increased if there are multiple collisions	3

6. O/N 25/P12/Q8/a

8(a)(i)	1 mark per bullet point, max 2 marks • Cloud storage can be free of charge for a limited amount • There is no need for personal storage devices • The data can be accessed from any computer with an internet connection • Cloud data services usually include data backup and recovery services • It is easier to share data / collaborate with colleagues • Cloud storage capacity can be quickly and easily increased if needed	2
8(a)(ii)	1 mark per bullet point, max 2 marks • The storage/cloud can only be accessed with an internet connection • There may be no backups // possibly no control over backups // reliance on third party for backups • Uploading / downloading data could take a long time • Potential issues with the cloud storage company impacting access e.g. downtime or technical problems • Potential compatibility issues if wanting to move the data • Limited storage availability without paying extra • ... meaning (possibly) more expensive in the long term • Possibly security issues	2

7. O/N 25/P13/Q4/a,b

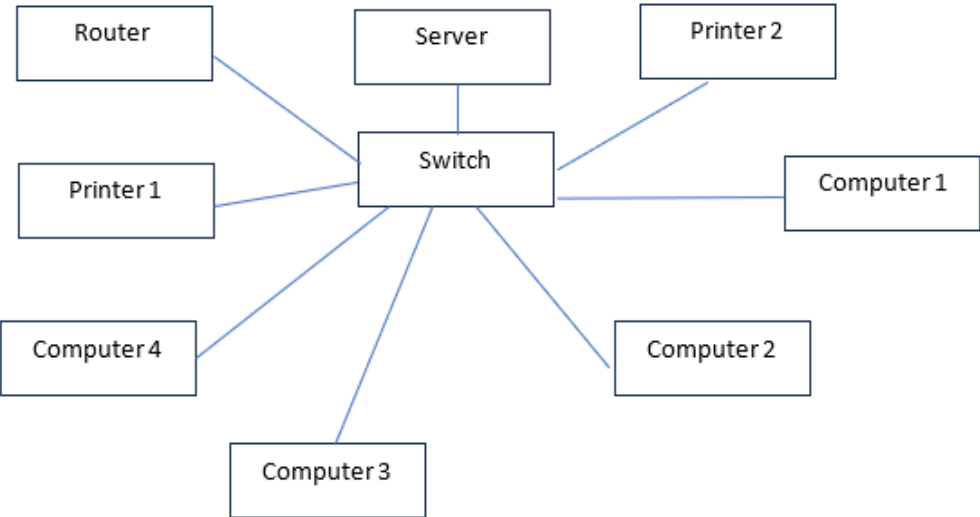
4(a)	1 mark for each benefit, max 3 • Devices can be more mobile as no physical connection is required • No cabling is needed so easier to set up • Easier to add additional devices • Multiple different device types can be connected	3
4(b)	1 mark for a drawback • Can be less secure • Transmission speed can be slow • Interference is possible • The signal degrades quickly without repeaters/boosters	1

8. M/J 25/P11/Q2/a,c

2(a)	1 mark each to max 4 • Video is transmitted continuously • as a series of bits • The video is uploaded to a media server • The users download from the media server • On download, the media server sends the data to a buffer on the user's device • buffer is used when there is a difference in speed between transmission and receipt • Buffer stores data from server until recipient can receive it • Recipient views bit stream from the buffer	4
2(c)	1 mark each modems: • Converts digital data into analogue for transmission down phone lines // Converts analogue data into digital after transmission down phone lines dedicated lines: • Used to provide a direct / private connection which therefore provides faster transmission	2

9. M/J 25/P12/Q6/a,b,c,d,e

6(a)	1 mark each to max 2 • In a WAN devices can connect over a larger geographical area in a LAN devices connect over a smaller geographical area • A WAN may use non-dedicated hardware the hardware in a LAN is usually dedicated • A WAN uses external / not company owned transmission media // the transmission media for a LAN is usually company owned	2
6(b)	1 mark each to max 4 e.g. • Land is split into cells designed for maximum line of sight • Each cell has a tower with an antenna which receives and transmits data • Data is transmitted between the tower and the phone • Data transmission is wireless using low power radio signals/frequencies • Multiple devices can communicate simultaneously with the same tower	4

6(c)	1 mark for each correct word • full stops • hexadecimal • zeros • 128 • dynamic • private IPv4 is made of four groups of 8-bit numbers separated by full stops. Each IPv4 address is 32 bits. IPv6 is made of eight groups of 4 hexadecimal numbers separated by colons. Multiple consecutive groups of zeros can be replaced with a double colon. Each IPv6 address is 128 bits. A dynamic IP address can change each time the computer connects to a network. A private IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.	6
6(d)	1 mark each • Server directly connected to the switch • Four labelled computers all connected directly to the switch • Two printers each directly connected to the switch / server / a computer • Router / modem directly connected to the switch / server for internet access e.g. Example only  <pre> graph TD Router[Router] --- Switch[Switch] Server[Server] --- Switch Printer1[Printer 1] --- Switch Printer2[Printer 2] --- Switch Computer1[Computer 1] --- Switch Computer2[Computer 2] --- Switch Computer3[Computer 3] --- Switch Computer4[Computer 4] --- Switch </pre>	4

6(e)	1 mark each to max 3 e.g. • To store the MAC addresses of devices connected to it • To receive packets from devices • To forward packets directly to the intended recipient • To provide a central point of connection for the network • to enable connected devices to communicate	3
------	--	----------

10. M/J 25/P13/Q3/c

3(c)	1 mark each to max 4 • A webserver stores all the data for each player • Each player is on a client computer // The player's web browser is the client • that sends requests over the internet to the web server • the server performs the required action in the game • the server updates the data in the game • the server sends the results to the player	4
------	--	----------

11. O/N 24/P13/Q9/a,b,c,d

9(a)	1 mark for each bullet point (max 2) • A WAN covers a large geographical area • External/public infrastructure is used • Non-dedicated hardware	2
9(b)	1 mark for correct media 1 mark for matching description e.g. • Fibre optic • Transmits data as pulses of light • Radio waves / microwaves • Transmits electromagnetic waves on different frequencies	4
9(c)	1 mark for each bullet point (max 4) • Data is compressed before transmitting • The video and audio are transmitted continuously as a series of bits • The video is uploaded to a media server • On download, the media server sends the data to a buffer • The buffer stores data from server until the receiving device can process it • The receiving device receives the bit stream from the buffer	4
9(d)(i)	1 mark for public IP address and 1 mark for private IP address: Public IP address: • So that the router is visible to the Internet/external network/WAN Private IP address: • So that the router is identified to computers within the LAN	2

9(d)(ii)	1 mark for each bullet point (max 2) • It allows the network to be divided into smaller networks • ... which reduces traffic in some parts of the network // reduces congestion • ... because traffic only travels through the parts necessary • ... which hides the complexity of network • ... and allows for easier maintenance of the network	2
----------	---	----------

12. M/J 24/P11/Q2/e

2(e)(i)	1 mark for: Continuous ordered flow of bits over a communication path	1
2(e)(ii)	1 mark each to max 2 : • Real-time is direct from source whereas on-demand is pre-recorded/downloaded to view later • Real-time cannot be re-watched, on-demand can be paused, re-watched etc. • Real-time plays continually, on-demand downloads sections/blocks and cannot play until next section is downloaded	2

13. M/J 24/P11/Q5/a

5(a)	1 mark each: • Identification of server in the bank scenario • Description e.g. Receives requests, processes the requests • Identification of client in bank scenario • Description e.g. Sends request to the server, waits and outputs the response	4
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14. M/J 24/P11/Q8/a,b

8(a)	1 mark for each correct row:	5																								
	<table><tr><th>Statement</th><th>Bus</th><th>Star</th><th>Mesh</th></tr><tr><td>all devices connect to one central device</td><td></td><td>✓</td><td></td></tr><tr><td>all devices connect to a central cable</td><td>✓</td><td></td><td></td></tr><tr><td>multiple paths for the packets to travel</td><td></td><td></td><td>✓</td></tr><tr><td>robust against damage because if any line fails, the rest of the network retains full functionality</td><td></td><td>✓</td><td>✓</td></tr><tr><td>most likely to lose data through collisions</td><td>✓</td><td></td><td></td></tr></table>	Statement	Bus	Star	Mesh	all devices connect to one central device		✓		all devices connect to a central cable	✓			multiple paths for the packets to travel			✓	robust against damage because if any line fails, the rest of the network retains full functionality		✓	✓	most likely to lose data through collisions	✓			
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most likely to lose data through collisions	✓																									
8(b)(i)	1 mark for: to be visible to and accessible by other devices on the internet	1																								

8(b)(ii)	1 mark each: - IPv4 has 4 groups of digits whilst IPv6 has 8 groups - IPv4 is usually represented in denary whilst IPv6 is usually represented in hexadecimal - IPv4 groups are between 0 and 255 whilst IPv6 is between 0 and FFFF - IPv4 is 32 bits whilst IPv6 is 128 bits	2
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15. M/J 24/P12/Q3/b,c

3(b)	1 mark for characteristic 1 mark for description of application to examination software: <table><tr><th>Thin-client characteristic</th><th>Description of use in this software</th></tr><tr><td>Data is not stored on the client computer</td><td>Exam papers are stored on the server and not on the examiner's computer // exam papers are not permanently stored on the examiners' computers</td></tr><tr><td>Client computer is reliant on access to server</td><td>Examiners cannot mark if their device cannot access the server / the server 'goes down'</td></tr><tr><td>Client computer heavily reliant on network/internet connection</td><td>The marking software will not operate without network/internet access</td></tr><tr><td>Client computer requires few local resources/memory</td><td>Examiners can use devices with low resources and the marking software will still function</td></tr><tr><td>Client computer performs minimal functions/processes</td><td>The marking software transmits requests, the server responds and sends the response to the user</td></tr></table>	Thin-client characteristic	Description of use in this software	Data is not stored on the client computer	Exam papers are stored on the server and not on the examiner's computer // exam papers are not permanently stored on the examiners' computers	Client computer is reliant on access to server	Examiners cannot mark if their device cannot access the server / the server 'goes down'	Client computer heavily reliant on network/internet connection	The marking software will not operate without network/internet access	Client computer requires few local resources/memory	Examiners can use devices with low resources and the marking software will still function	Client computer performs minimal functions/processes	The marking software transmits requests, the server responds and sends the response to the user	4
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Client computer performs minimal functions/processes	The marking software transmits requests, the server responds and sends the response to the user													
3(c)(i)	1 mark each to max 2: • Receives packets from internet • Analyses the destination IP address of each packet • Forwards the packet towards its destination • ... using the routing table • Maintains/updates the routing table • Finds the most efficient route to the destination	2												
3(c)(ii)	1 mark each to max 2: • The PSTN consists of many different types of communication lines • ... therefore the digital data may need to be converted into a different form/analogue signal • Data is transmitted in both directions at the same time // duplex data transmission • Using a PSTN the communication passes through different switching centres/ISPs	2												

16. M/J 24/P13/Q5/a,b,c,d

5(a)(i)	1 mark for: Dedicated/bespoke services/storage on a remote server only available to company	1
5(a)(ii)	1 mark each to max 3: e.g. • Not reliant on a third party • ... gives greater control over security/privacy • ... gives greater control over backup • Storage can be tailored/scalable to company requirements// an example e.g. the amount of storage accessible/ facilitating the sharing of files	3
5(b)	1 mark each: • Sending computer transmits packets directly to switch/router/central device • Switch/router/central device checks destination address of packet and forwards directly to that device	2
5(c)(i)	1 mark each to max 2: • Jamming signal is transmitted by the sending device • Transmission is aborted • The sending device waits a random time before trying to send data again ... • ... if further collisions occur the wait time is increased	2
5(c)(ii)	1 mark each to max 2: • Random time increased each time so can be infinite waiting • May be constant jamming signal so nothing ever sends • Certain nodes cannot be prioritised • High power consumption • Only suitable for short distance network // limited distance • Not scalable // more nodes means exponentially longer waiting times	2
5(d)	1 mark for: Static means the IP for that device does not change and Private means it can only be accessed/seen/used within the LAN	1

17. O/N 23/P11/Q2/a,b,c,d

2(a)	1 mark for each bullet point (max 3) • Receives packets from internet / external network • Implements a firewall • Analyses the destination IP address of each packet • Forwards the packet towards its destination // send packets onto local network or external network • ...using the routing table • Maintains / updates the routing • Allocates <u>private</u> IP addresses • Finds the most efficient route to the destination • Changes the packet format for transmission over the next network // • Network Address Translation (NAT): NAT is a technique used by routers to allow multiple devices in a private local area network (LAN) to share a single public IP address.	3
2(b)	Switch: 1 mark for each bullet point (max 1) • To allow two or more devices to communicate with one another • To connect individual devices to each other • To receive transmissions and forward them to their destination Wireless Access Point (WAP): 1 mark for each bullet point (max 1) • To allow connection of devices (to the central device) using radio signals / Wi-Fi • To allow the central device to send / receive radio signals / Wi-Fi signals • To allow wireless enabled devices to connect to a wired network Bridge: 1 mark for each bullet point (max 1) • To connect two LANs / segments with the same protocol • To transmit data between two networks with the same protocol	3
2(c)	1 mark for each bullet point (max 2) • The students cannot access their files without a reliable internet connection • The amount of space for no payment may be limited so students will have to purchase more space if needed • The students do not have control over the backup (or security) of their work // the students are dependent on a third party for the (security and) backing up of their work	2

2(d)	1 mark for each advantage and 1 mark for valid corresponding expansion - Star topology is more resilient to faults ...because there is no single cable and leads to less disruption to teaching - Higher performance as fewer collisions ...because each device in the classroom is only connected to the switch - Easier to add new nodes ...because each device in the classroom connects directly to the switch - Easier to fault find compared to bus topology	2
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18. O/N 23/P12/Q7/a,b,c,d

7(a)	1 mark for each bullet point (max 2) - Only has four groups of digits // IPv6 has eight groups - Uses dotted notation instead of colons - Because it is a 32 bit / 4 byte address // IPv6 is 128 bits / 16 bytes	2
7(b)(i)	1 mark for server only connected to router 1 mark for two laptop computers connected only to router <pre> graph TD Server[Server] --- Router[Router] Router --- Laptop1[Laptop 1] Router --- Laptop2[Laptop 2] </pre>	2
7(b)(ii)	1 mark for each bullet point (max 2) - The data from the sending laptop is transmitted to the router - The data has address of recipient - The router determines recipient's destination address ... by using a routing table - The router transmits data directly / only to recipient	2

7(b)(iii)	1 mark for each use (max 2) and 1 mark for corresponding expansion (max 2) • To improve the security of the LAN • ... so that devices do not receive unintended data • ... so that a compromised device does not expose the whole network • ... so not all devices can access all segments • To make the network management easier • ... because faults can be isolated more efficiently • ... by appropriate example • To make the network easier to expand // For better control of network growth • ... by allowing for greater range of IP addresses to be available • To improve network performance • To reduce network congestion • ... by localising network communications // by dividing data between segments • ... so that devices are not flooded with data • ... because data sent between devices on the same subnet stays within the subnet	4										
7(c)	1 mark for each bullet point (max 3) • To monitor the communications channel • To send data only when there is no data being transmitted / the line is quiet / idle • To detect a collision and then stop transmissions of further data // transmit a jamming signal • To calculate a random wait time / back-off time • ... then retransmit the data after that random wait time • Increase random time if multiple collisions	3										
7(d)	1 mark for each highlighted area <table><tr><th>Type of IP address</th><th>Description</th></tr><tr><td>public IP address</td><td>an IP address that is assigned to a device to allow direct access on the internet</td></tr><tr><td>static IP address</td><td>an IP address that is fixed / doesn't change each time a device re-joins a LAN / network</td></tr><tr><td>private IP address</td><td>an IP address used for internal LAN communication only</td></tr><tr><td>dynamic IP address</td><td>an IP address that may be refreshed / changed each time a device re-joins a LAN / network</td></tr></table>	Type of IP address	Description	public IP address	an IP address that is assigned to a device to allow direct access on the internet	static IP address	an IP address that is fixed / doesn't change each time a device re-joins a LAN / network	private IP address	an IP address used for internal LAN communication only	dynamic IP address	an IP address that may be refreshed / changed each time a device re-joins a LAN / network	4
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19. O/N 23/P13/Q3/a(i,ii)

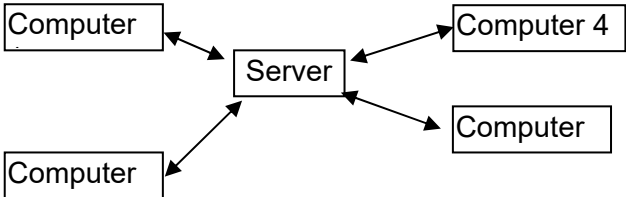
3(a)(i)	1 mark for each bullet point (max 2). • Courses must be available to anyone who wishes to follow them • Courses must be available on the internet • Company is willing to share infrastructure with other companies (public) • ...which is more economic for the company	2
3(a)(ii)	1 mark for each bullet point (max 2 for each disadvantage). • There could be a possible loss of control unlike the LAN • ...because the data is stored on a remote infrastructure / someone else's infrastructure • ...reliance on external agency to complete tasks, e.g. backups, security • Requires reliable internet connection • ...to ensure access to the remote data, more likely with LAN • Increased recurring costs • ...as cloud provider charges must be paid, costs for LAN once only.	4

20. M/J 23/P11/Q4/d

4(d)(i)	1 mark for each term An IPv4 address contains 4 groups of digits. Each group is represented in 8 bits and the groups are separated by full stops. An IPv6 address contains 8 groups of digits. Each group is represented in 16 bits. Multiple groups that only contain zeros can be replaced with a :: // double colon .	5
4(d)(ii)	1 mark for identification, 1 mark for expansion e.g. • Reduce amount of traffic in a network // improve network speed • Data stays in its subnet so it does not travel as far • Improves network security • .. so that not all devices can access all areas of the network • Allows for easier maintenance • ... because only one subnetwork may need taking down/changing while the rest of the network can continue	4

21. M/J 23/P12/Q1/a,b,c,d,e

1(a)	1 mark each to max 2 • Allows the sharing of files/data // Allows communication between the devices • Allows the sharing of resources e.g. hardware / software (applications) • Allows central management // by example, backup, security, etc.	2
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1(b)	1 mark each to max 2 - Covers a small geographical area - The infrastructure is privately owned // not controlled by external organisations	2
1(c)	1 mark each to max 2 - Each computer directly connected only to the server <u>all</u> components correctly labelled  <pre> graph TD S[Server] <--> C1[Computer] S <--> C2[Computer 4] S <--> C3[Computer] S <--> C4[Computer] </pre>	2
1(d)	1 mark each to max 3 - A protocol (suite) - For data transmission over standard / universal wired / cabled network connections - Uses Carrier Sense Multiple Access / Collision Detection (CSMA/CD) - Data is transmitted in frames ... each frame has a source and destination (IP/MAC) address ... and error checking data (so damaged frames can be resent)	3
1(e)	1 mark each - The server performs minimal / some processing for the client - The clients also do most of their own processing/work independently // most of the resources are installed locally	2

22. M/J 23/P13/Q2/a,b,c,d,e

2(a)	1 mark each to max 2 - WAN covers a large geographical area and LAN covers a small geographical area - LAN connections between devices are usually physical, whereas the WAN connections are often virtual - A LAN has a high data transfer rate, whereas a WAN has a low data transfer rate - The ownership of a LAN is private; the ownership of a WAN can be private or public - LAN is usually more secure than a WAN because protection is easier to implement	2
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2(b)(i)	1 mark each to max 2 - All computers are connected to at least one other device - There are multiple routes between devices - The computers can act as relays, passing packets on towards the final destination	2
2(b)(ii)	1 mark each to max 2 - If one line goes down there are more routes available - Improved security as not using one main line - No/fewer collisions - New nodes can be added without interruption or interfering with other nodes - More secure because data is sent over a dedicated connection	2
2(c)	1 mark each to max 2 - Server performs all processes required by the task and/or data storage - Clients only sends requests to the server and displays the returned results	2
2(d)	1 mark each to max 4 Examples: - Some students might only have one sort of connection on their device - Wired provides better performance for the student's device ... for example, enabling faster access to university databases - There will be less interference if students connect via a cable - Students can transmit private/confidential data/work securely ... for example, their final dissertation - Wireless connection means that the students can use their devices in different rooms/sites/outside/anywhere more freely // student devices can be portable - Wireless connection enables the students to bring multiple devices // bring their own devices // change devices	4
2(e)	1 mark for identification - IP address is made up of a <u>network ID</u> and a <u>host ID</u> 1 mark each to max 2 for description - Each device in a subnetwork has the same network ID // Each subnetwork has a different network ID - Every device in each subnetwork has a different host ID but the same network ID // the host ID uniquely identifies the device within the subnetwork	3

23. O/N 22/P11/Q8

8	1 mark for bullet point (max 4): - CSMA/CD is a protocol used to detect and prevent collisions in a bus topology - before transmitting, a device checks if the channel is busy - If the channel is busy the device waits // if the channel is free the data is sent - because there is more than one computer connected to the same transmission medium ... two workstations can start to transmit at the same time, causing a collision - If a collision is detected by the device, transmission is aborted / a jamming signal is transmitted - both devices wait a (different) random time and then try again	4
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24. O/N 22/P12/Q10/a,b

10(a)	all four computers directly connected to the switch and no other connections.	1
10(b)(i)	1 mark for the device. 1 mark for corresponding reason. - Device: Server - Reason: Server processes the requests and authorises traffic // firewall software on the server authorises traffic // server acts as the proxy - Device: Switch - Reason: Switch is connected to all the computers // to share access to the router on the network	2
10(b)(ii)	1 mark for each bullet point (max 3): - receive packets from devices / internet - find destination of packets using the IP address - forward packets to the destination - assign private IP addresses to devices on LAN - store/update/maintain a routing table - find most efficient path to destination - maintain table of MAC and IP addresses - provides the LAN with a public IP address - acts as a gateway - performs protocol conversion - acts as a firewall	3

25. O/N 22/P13/Q7/a,b,c

7(a)	1 mark for each correct answer: <table><tr><th colspan="2">Answer</th></tr><tr><td>The name of device A that allows the laptop to connect to the internet</td><td>Router</td></tr><tr><td>A type of cloud, X</td><td>Public (cloud)</td></tr><tr><td>An example of an application, B, that can run on the cloud,</td><td>Email / Graphics / Word processor / Spreadsheet / Game / Database, etc.</td></tr></table>	Answer		The name of device A that allows the laptop to connect to the internet	Router	A type of cloud, X	Public (cloud)	An example of an application, B, that can run on the cloud,	Email / Graphics / Word processor / Spreadsheet / Game / Database, etc.	3
Answer										
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A type of cloud, X	Public (cloud)									
An example of an application, B, that can run on the cloud,	Email / Graphics / Word processor / Spreadsheet / Game / Database, etc.									
7(b)	1 mark for a correct advantage: - not fixed to a single location - allows access in remote / rural areas 1 mark for each correct disadvantage (max 2): - high latency / lag / slow to connect - more expensive than wired methods, as need extra equipment - signal is affected by bad weather - the transmission speed is slower than fixed line broadband - direct line of sight needed	3								
7(c)(i)	1 mark for each bullet point (max 2) Examples: - improves security - reduces congestion - allows extension of the network / devices attached - aids day-to-day management - improves performance	2								
7(c)(ii)	1 mark for each correct answer: - network ID = 10 - host ID = 4	2								

26. O/N 21/P11/Q8/a,b,c

8(a)	1 mark per bullet point - LAN - Small geographical area - No leasing external infrastructure / transmission media // does not use internet to transmit within the building	3
8(b)	1 mark per item - router - switch - hub	2

8(c)	1 mark per bullet point to max 4 • Provide interface to wireless network • ... as an antenna • Receives analogue radio waves • ... convert them to digital / binary • Checks incoming transmissions for correct MAC / IP address • ... ignore transmissions not intended for it • Encrypts / encodes the data • Decrypts / decodes the data • Takes digital/binary input and converts to analogue waves • ... sends the radio waves via the antenna	4
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27. O/N 21/P12/Q3/b

3(b)(i)	1 mark for identification of star topology 1 mark for justification Devices are connected directly to the <u>router</u> independently // all devices are only connected to the <u>router</u>	2
3(b)(ii)	1 mark for each correct function to max 3 • To receive packets from devices or the Internet • To forward / route packets to the destination • To find the destination of the packet • To assign / allocate private IP addresses to devices on LAN • To store / update / maintain a routing table • To find the most efficient path to the destination • To maintain a table of MAC and IP addresses	3

28. O/N 21/P13/Q8/a,b,c

8(a)	1 mark per bullet point • LAN • Small geographical area • No leasing external infrastructure / transmission media // does not use internet to transmit within the building	3
8(b)	1 mark per item • router • switch • hub	2

8(c)	1 mark per bullet point to max 4 • Provide interface to wireless network • ... as an antenna • Receives analogue radio waves • ... convert them to digital / binary • Checks incoming transmissions for correct MAC / IP address • ... ignore transmissions not intended for it • Encrypts / encodes the data • Decrypts / decodes the data • Takes digital/binary input and converts to analogue waves • ... sends the radio waves via the antenna	4
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29. 9608 O/N 21/P11/Q11/a,b

11(a)	One mark per bullet point to max 2 • Web pages/files are saved on servers • Clients send requests to the web servers • Web servers process the requests • ...and return the results to the client • ... client displays the results to the user	2
11(b)	One mark per bullet point to max 3 IPv4 • Four groups of (denary or Hexadecimal integers) • Numbers between 0 and 255 / 0 and FF • Each stored in 1 byte / 8 bits // the whole is stored in 32 bits / 4 bytes • Separated by full stops • Correct example OR IPv6 • Eight groups of (Hexadecimal) digits • Numbers between 0 and FFFF • Each stored in 2 bytes/16 bits // the whole stored in 128 bits / 16bytes • Separated by colons • The first instance of multiple groups of zero can be replaced by a double colon • correct example	3

30. M/J 21/P11/Q4/a,b,c,d

4(a)	1 mark per bullet point to max 2 • All computers are of equal status • Each computer provides access to resources and data // data is distributed • Computers can communicate and share resources • Each computer is responsible for its own security	2
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4(b)	1 mark per bullet point to max 2 per drawback • Reduced security // no central management of security • ... only as secure as the weakest computer on the network • ... each computer is at risk from viruses from other computers • No central management of backup • ... if the data from one computer is not backed up it is lost to all of them • No central management of files/software • ... consistency may be difficult to maintain • ... each computer may have different software from the others • Individual computers may respond slower • ... because they are being accessed by other computers • In order to share files etc. all the computers involved need to be switched on • ... so the files etc. may not be always available	4															
4(c)(i)	1 mark for first 2 ticks, 1 mark for last 2 (shaded) <table border="1"> <thead> <tr> <th>Task</th><th>Performed by router</th><th>Not performed by router</th></tr> </thead> <tbody> <tr> <td>Receives packets from devices</td><td>✓</td><td></td></tr> <tr> <td>Finds the IP address of a Uniform Resource Locator (URL)</td><td></td><td>✓</td></tr> <tr> <td>Directs each packet to all devices attached to it</td><td></td><td>✓</td></tr> <tr> <td>Stores the IP and/or MAC address of all devices attached to it</td><td>✓</td><td></td></tr> </tbody> </table>	Task	Performed by router	Not performed by router	Receives packets from devices	✓		Finds the IP address of a Uniform Resource Locator (URL)		✓	Directs each packet to all devices attached to it		✓	Stores the IP and/or MAC address of all devices attached to it	✓		2
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4(c)(ii)	1 mark per bullet point for justification up to max 3 No mark for identification of wired/wireless Wired • Faster connection // higher bandwidth • needed as she is downloading/streaming large files • ... less time waiting / less latency / fewer delays • More reliable / stable connection • ... is less susceptible to issues with distance/walls/interference • More secure Wireless • Freedom of movement • ... can move between different rooms with a mobile device and still receive/transmit data • ... no need of a physical connection • Easily expanded if friends want to access the same network • Less cabling / expertise is needed • ... making the initial setup less expensive	3
4(d)	1 mark for identifying that she is using both. 1 mark per bullet point for justification • using internet because sending data on the infrastructure • using WWW because accessing a website (that is stored on a web server operated by the webmail) that is part of the WWW	3

31. M/J 21/P12/Q5/c,d

5(c)(i)	1 mark per bullet point to max 2 • Cloud storage can be free (for small quantities) • No need for separate (high capacity) storage devices // saves storage on existing devices • Can access data from any computer with internet access • Most cloud data services will have in-built backup/disaster recovery • Security could be better • Can easily increase capacity • Data can be easily shared	2
5(c)(ii)	1 mark per bullet point to max 2: • Can only access (the cloud) with internet access • Security may not be strong // no control over security • There may not be any backups // no control over backups • It can take a long time to upload/download the data • It can be more expensive in the long term • There could be a limit to the amount of storage unless paid for • There could be compatibility/access issues • There could be issues with the company offering cloud services	2

5(d)

1 mark for each correct line

4

Term	Description
Public IP Address	It is only visible to devices within the Local Area Network (LAN)
Private IP address	It increments by 1 each time the device connects to the internet
Dynamic IP address	A new one is reallocated each time a device connects to the internet
Static IP address	It can only be allocated to a router
	It is visible to any device on the internet
	It does not change each time a device is connected to the internet

32. M/J 21/P13/Q4/a,b,c,d

4(a)	1 mark per bullet point to max 2 • All computers are of equal status • Each computer provides access to resources and data // data is distributed • Computers can communicate and share resources • Each computer is responsible for its own security	2
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4(b)	1 mark per bullet point to max 2 per drawback • Reduced security // no central management of security • ... only as secure as the weakest computer on the network • ... each computer is at risk from viruses from other computers • No central management of backup • ... if the data from one computer is not backed up it is lost to all of them • No central management of files/software • ... consistency may be difficult to maintain • ... each computer may have different software from the others • Individual computers may respond slower • ... because they are being accessed by other computers • In order to share files etc. all the computers involved need to be switched on • ... so the files etc. may not be always available	4															
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4(d)	1 mark for identifying that she is using both. 1 mark per bullet point for justification • using internet because sending data on the infrastructure • using WWW because accessing a website (that is stored on a web server operated by the webmail) that is part of the WWW	3

33. 9608 M/J 21/P11/Q9/a,b

9(a)	1 mark per difference • Private IP is only known within the LAN // Public IP is known outside of the LAN/ on Internet • Public is allocated by ISP // Private is allocated by the router • Public addresses are unique throughout the Internet, private addresses are unique only within the LAN • Private IP addresses are more secure than public IP addresses	2										
9(b)	1 mark for each correct term <table><tr><th>Description</th><th>Term</th></tr><tr><td>Receives data packets from a network and forwards them onto a similar network</td><td>Router</td></tr><tr><td>Manages access to a centralised resource</td><td>Server</td></tr><tr><td>Joins networks that use different sets of rules to transmit data</td><td>Gateway</td></tr><tr><td>Monitors and controls incoming and outgoing network traffic based on set criteria</td><td>Firewall</td></tr></table>	Description	Term	Receives data packets from a network and forwards them onto a similar network	Router	Manages access to a centralised resource	Server	Joins networks that use different sets of rules to transmit data	Gateway	Monitors and controls incoming and outgoing network traffic based on set criteria	Firewall	4
Description	Term											
Receives data packets from a network and forwards them onto a similar network	Router											
Manages access to a centralised resource	Server											
Joins networks that use different sets of rules to transmit data	Gateway											
Monitors and controls incoming and outgoing network traffic based on set criteria	Firewall											

34. 9608 M/J 21/P12/Q9/a,b

9(a)	1 mark for identification, 1 mark for description (max 2) Benefits: • Less hard-wiring/hardware is required • Users and computers can be mobile • It is much more straightforward to connect other devices Descriptions: • Reduced cost of setting up the network • The network can be accessed from anywhere within range of an access point • No need to physically connect each device	2
9(b)	1 mark for identification, 1 mark for matching description (max 2 for each) • Transmissions may be less secure • ...because data packets can be intercepted // easier connection by unauthorised user • Bandwidth may be limited // As more devices connect the bandwidth can be reduced • ...so access may be slow • It is subject to interference from other signals or obstacles • ...which can hinder transmission or corrupt data • Limited range // greater attenuation • ...so there is a need for repeaters // users can easily move out of range • Higher latency • ...so transmission will be slower	4

35. 9608 M/J 21/P13/Q6/b

6(b)	1 mark per bullet point for justification to max 3 • IP address is (physical) address of the server • A domain name is a memorable form for an IP address • <code>cambridgeinternational.org</code> is the domain name • An example of an IP address is 198.162.2.1 • Each domain name is linked to an IP address • A Domain Name Server/Service (DNS) is used to translate a domain name into its corresponding IP address • Domain name does not change but IP address could change (dynamic) • IP address can be used to address server directly, • The domain name needs translation before server can be assessed	3
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36. M/J 20/P12/Q7/a,b

7(a)	1 mark for device, 1 mark for matching purpose. Max 2 devices. - Router - To connect (devices) / the servers to the Internet // to transmit data between the servers and the Internet // to forward data towards its destination - Gateway - To connect a server that uses a different protocol to the Internet // to join two different types of network - Modem - To connect (the servers) to the Internet over a telephone line - Network interface card // NIC - To enable the servers to connect to the (company) network	4
7(b)	1 mark per bullet point, max 1 for benefit, max 1 for drawback Benefit: (Consistently) faster data transmission - More stable connection Drawback: - High initial cost as new hardware will be needed - Expertise required to complete connections	2

37. M/J 20/P13/Q1/a,b,c

1(a)	1 mark for each correctly completed media or description	4										
<table><tr><th>Type of connection</th><th>Description</th></tr><tr><td>Fibre-optic</td><td>Transmits data as light // Uses (a bundle of) glass/plastic threads to transmit data</td></tr><tr><td>Satellite</td><td>A communication device in Earth's orbit that receives and transmits data</td></tr><tr><td>Radio Waves</td><td>Carries data wirelessly, often known as Wi-Fi // Carries data in the form of electromagnetic waves</td></tr><tr><td>Copper cable</td><td>Carries data as electrical signals and can consist of a twisted pair</td></tr></table>			Type of connection	Description	Fibre-optic	Transmits data as light // Uses (a bundle of) glass/plastic threads to transmit data	Satellite	A communication device in Earth's orbit that receives and transmits data	Radio Waves	Carries data wirelessly, often known as Wi-Fi // Carries data in the form of electromagnetic waves	Copper cable	Carries data as electrical signals and can consist of a twisted pair
Type of connection	Description											
Fibre-optic	Transmits data as light // Uses (a bundle of) glass/plastic threads to transmit data											
Satellite	A communication device in Earth's orbit that receives and transmits data											
Radio Waves	Carries data wirelessly, often known as Wi-Fi // Carries data in the form of electromagnetic waves											
Copper cable	Carries data as electrical signals and can consist of a twisted pair											
1(b)	1 mark per bullet point to max 2 • The employees' computers are the clients • The server hosts the (shared) files • An employee can request a file (from the server) from any of the client computers • Several employees can access the same file (on the server) at the same time	2										

1(c)	1 mark per bullet point - Improved security because the IP address is not visible outside the network - An internet presence is not required for each employee computer - Only the router needs a public IP address, as only the router needs to be externally visible - Reduces number of (public) IP addresses needed	2
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38. O/N 19/P11/Q4/a,c

4(a)(i)	1 mark per bullet point to max 3 . If no application to the bank max 2 - The bank's server holds the customer account data / website ... and performs the requested tasks / processes. The computers used by the customers are the clients that send requests to the server ... which returns the results of the request ... E.g. a customer asks for a list of recent transactions on their account.	3
4(a)(ii)	1 mark per example to max 2 e.g. - Sending and receiving email - A company or school centrally storing files - Using a print server - Using a file server	2
4(c)(i)	1 mark per bullet point to max 2 - Less interference in the signal - The signal does not degrade as quickly // Needs less signal boosting - More secure // more difficult to hack - Greater bandwidth // <u>Faster</u> transmission speeds possible	2
4(c)(ii)	1 mark per bullet point to max 2 - Initial installation cost is higher // Cable / hardware is more expensive to buy per metre - Specialists / trained personnel are needed to install / maintain - Difficult to terminate // The electronics at both ends are more complex - Fibres can break <u>when bent</u> - Only transmits data in one direction // Cannot transmit power, only data	2

39. O/N 19/P12/Q6/e(i,ii)

6(e)(i)	1 mark per bullet to max 3 - The data is compressed before transmitting - The video is transmitted continuously as a series of bits - The video is hosted on a media server - On download, the server sends the data to a buffer on the client computers // The buffer stores the data from the server - The recipient / user's software receives bit stream from the buffer	4
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6(e)(ii)	1 mark for: On-demand 1 mark for justification from: - The video does not need to be broadcast live // the video is already recorded - Dominic's colleagues will watch the video at a later date // at their convenience	2
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40. O/N 19/P13/Q1/c(i)

1(c)(i)	1 mark per bullet point to max 3 plus 1 mark for suitable example - When a barcode on an item is scanned ... the server performs any requested tasks // the server looks up the details of the product - The self-checkout machine is a client ... that send requests to the server // the self-checkout machine asks for, e.g. the price of the item - The server returns the results of the request // the server returns e.g. the item price - Self-checkout machine displays e.g. price to the user	4
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41. M/J 19/P11/Q1/a,b,c

1(a)	1 mark for each correct indication and explanation 3A.21.2H.1 Invalid H is not a valid hexadecimal digit 299.53.2.2 Invalid 299 is not in the correct range 192.2.1.0 Valid Consists of four numbers in the range 0–255 separated by full stops	3
1(b)	1 mark per bullet point to max 3 - URL is parsed to obtain the Domain name - Domain name is sent to the nearest Domain Name Server (DNS) - DNS holds a list of Domain names and matching IP addresses - DNS name resolver searches its database for the Domain name - If DNS does not find the Domain name, the request is forwarded to a higher level DNS - If the Domain name is found, the IP address is returned - If the Domain name is not found, the request is passed to a higher level server - If the Domain name is finally not found, an error message is generated	3
1(c)	1 mark for each correct term Real-time On-demand	2

42. M/J 19/P12/Q1/a,b,d

1(a)(i)	1 mark for any valid example e.g. 192.168.0.1	1
1(a)(ii)	1 mark for correct answer The number of IP addresses needed will exceed the number available using IPv4.	1
1(a)(iii)	1 mark per bullet point to max 2 • Too many digits per group • Too many groups of digits • The address is more than 32 bits / 4 bytes • Colons are used as separators	2
1(b)(i)	1 mark per bullet point to max 2 • The PSTN consists of many different types of communication lines • Data is transmitted in both directions <u>at the same time</u> // (full) <u>duplex</u> data transmission • The communication passes through different switching centres	2
1(b)(ii)	1 mark for benefit, 1 mark for drawback Benefit • (Probably) faster connection / communication / transmission of data • (Usually) more consistent transmission speed • Improved security Drawback • Expensive to <u>set-up / maintain</u> • Disruption to the dedicated line would leave no alternative	2
1(d)	1 mark per bullet point to max 3 for any valid answer For example: • File server • Print server • Proxy server • Web server • Application server	3

43. O/N 18/P11/Q2/a,b,c

2(a)	Use the IP address instead of the URL	1
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2(b)(i)	1 mark per correct answer <table><tr><th>IP Address</th><th>Valid or invalid</th></tr><tr><td>21E5:69AA:FFFF:1:E100:B691:1285:F56E</td><td>Valid</td></tr><tr><td>::255.255.255.255</td><td>Valid</td></tr><tr><td>59FB::1005:CC57:6571</td><td>Valid</td></tr><tr><td>56FE::2159:5BBC::6594</td><td>Invalid</td></tr></table>	IP Address	Valid or invalid	21E5:69AA:FFFF:1:E100:B691:1285:F56E	Valid	::255.255.255.255	Valid	59FB::1005:CC57:6571	Valid	56FE::2159:5BBC::6594	Invalid	4					
IP Address	Valid or invalid																
21E5:69AA:FFFF:1:E100:B691:1285:F56E	Valid																
::255.255.255.255	Valid																
59FB::1005:CC57:6571	Valid																
56FE::2159:5BBC::6594	Invalid																
2(b)(ii)	1 mark per correct row <table><tr><th>Statement</th><th>Public</th><th>Private</th></tr><tr><td>192.168.2.1 is an example of this type of address</td><td></td><td>✓</td></tr><tr><td>Assigned by the Internet Service Provider (ISP)</td><td>✓</td><td></td></tr><tr><td>IP address cannot be duplicated in different networks</td><td>✓</td><td></td></tr><tr><td>Network Address Translation (NAT) is necessary to access the Internet directly</td><td></td><td>✓</td></tr></table>	Statement	Public	Private	192.168.2.1 is an example of this type of address		✓	Assigned by the Internet Service Provider (ISP)	✓		IP address cannot be duplicated in different networks	✓		Network Address Translation (NAT) is necessary to access the Internet directly		✓	4
Statement	Public	Private															
192.168.2.1 is an example of this type of address		✓															
Assigned by the Internet Service Provider (ISP)	✓																
IP address cannot be duplicated in different networks	✓																
Network Address Translation (NAT) is necessary to access the Internet directly		✓															
2(c)	1 mark per example to max 2 - Fibre-optic - Wi-Fi / Radio waves - Microwave - Infrared	2															

44. O/N 18/P12/Q2/a,b,c

2(a)	1 mark for 1 correct answer, 2 marks for all 3 correct answers - Gopal types into the web browser B (Web browser sends URL to Domain name Service (DNS)) - DNS looks up URL in a table A (DNS finds corresponding IP address) C (DNS returns IP address to web browser)	2
2(b)	1 mark per bullet point to max 2 - Gives each device on a network an identifier // IP address used to locate a device on a network - Each address is <u>unique</u> within the network - Allows a device/gateway/node to send data to the correct destination / a specific device/gateway/node	2

2(c)(i)	1 mark per bullet point to max 3 - Less interference in signal - Signal does not degrade as fast // Needs less signal boosting - More difficult to hack // more secure - Greater bandwidth // <u>Faster</u> transmission speeds possible	3
2(c)(ii)	1 mark per bullet point to max 2 (Initial) installation cost is higher // Cable / hardware is more expensive to buy (per metre) - Specialists / trained personnel needed to install / maintain - Difficult to terminate // Electronics at both ends are more complex - Fibre-optic cables can break <u>when bent</u> - Only transmits data in one direction - If a fibre-optic cable connection fails, many more services can be affected	2

45. O/N 18/P13/Q4/a,b,c

4(a)	1 mark for 1 letter in correct space 2 marks for all 3 letters in correct places 1 C 2 URL goes to Domain Name Service (DNS) 3 B 4 A 5 DNS returns IP address to client	2															
4(b)(i)	1 mark per bullet point <u>258</u> is too large/largest individual numbers is 255 4 numbers needed/1 number missing/only 3 groups of numbers given	2															
4(b)(ii)	1 mark per bullet point L not a valid hexadecimal number - Only one double colon is allowed	2															
4(c)	1 mark per row <table border="1" data-bbox="300 1637 1326 1995"> <thead> <tr> <th>Description</th><th>Public</th><th>Private</th></tr> </thead> <tbody> <tr> <td>The address can be reached over the Internet.</td><td>✓</td><td></td></tr> <tr> <td>The address is more secure.</td><td></td><td>✓</td></tr> <tr> <td>The address can only be accessed through the same LAN.</td><td></td><td>✓</td></tr> <tr> <td>The address can be duplicated in different networks.</td><td></td><td>✓</td></tr> </tbody> </table>	Description	Public	Private	The address can be reached over the Internet.	✓		The address is more secure.		✓	The address can only be accessed through the same LAN.		✓	The address can be duplicated in different networks.		✓	4
Description	Public	Private															
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The address is more secure.		✓															
The address can only be accessed through the same LAN.		✓															
The address can be duplicated in different networks.		✓															

46. O/N 18/P13/Q7/a,b

7(a)	1 mark per benefit to max 3 • Devices can be more mobile as they do not have to be connected to cable • Easier to set up // no cables need to be installed • Add additional devices is easier • Many different types of device can be connected at the same time	3
7(b)	1 mark for a drawback from the following: • <u>Easier</u> to hack • Interference • Signal degrades quickly	1

47. M/J 18/P11/Q1

1	1 mark for a correct line from each communication media, max 6	6
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48. M/J 18/P11/Q5/a,b

5(a)	1 mark per server e.g. • E-mail • Print • Web	2
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5(b)	1 mark for the indicating the statement is false: <table><tr><td>True</td><td>False</td></tr><tr><td></td><td>✓</td></tr></table> 1 mark per bullet for justification to max 4. • Internet is the infrastructure / global collection of networks • World Wide Web is the (multimedia web) pages / content • The World Wide Web is accessed over the Internet • Webpages are written in HTML • HTTP protocol used to transfer web pages • Internet uses IP protocol	True	False		✓	5
True	False					
	✓					

49. M/J 17/P12/Q6/a,b

6(a)	Two from: • The <u>file</u> is made available from a web/email/FTP server 1 • The user's <u>browser</u> is the client software 1 • The client (software browser) <u>requests</u> the <u>file</u> from the server 1 • The desired <u>file</u> is returned to the client computer 1	Max 2
6(b)	1. The user keys in the Uniform Resource Locator (URL) into the browser Software. 2. E // The Domain Name Service (DNS) uses the domain name from the browser to look up the IP address of the web server. 1 3. D // The web server retrieves the page 1 4. F // Sends the web page content to the browser 1 5. B // Browser software renders the page and displays 1	4

50. O/N 16/P11/Q6

6 Any **four** from:

- User needs high-speed broadband (connection)
- Data is streamed to a buffer (in the computer)
- Buffering stops video pausing as bits streamed
- As buffer is emptied, it fills up again so that viewing is continuous
- Actual playback is (a few seconds) behind the time the data is received by computer [4]

51. O/N 16/P13/Q6

6 Any **four** from:

- User needs high-speed broadband (connection)
- Data is streamed to a buffer (in the computer)
- Buffering stops video pausing as bits streamed
- As buffer is emptied, it fills up again so that viewing is continuous
- Actual playback is (a few seconds) behind the time the data is received by computer [4]

52. M/J 16/P11/Q7/a,b,c**7 (a) Internet Protocol****[1]****(b)****[4]**

Address	Denary / Hexadecimal	Valid or Invalid	Reason
3.2A.6AA.BBBB	Hexadecimal	Invalid	One point from: - This is more than <u>32 bits</u> <u>6AA / BBBB</u> in Hex is bigger than <u>FF / 255</u> in denary <u>6AA / BBBB</u> uses more than 8 bits / a byte - The third / fourth group is bigger than <u>FF / 255</u> in denary - The third / fourth group uses more than 8 bits / a byte
2.0.255.1	Denary	Valid	There are 4 bytes, each 255 or below // All the values are in the range 0 - 255
6.0.257.6	Denary	Invalid	<u>257</u> is above 255 // The third group is above 255
0A.78.F4.J8	Hexadecimal	Invalid	J is not a valid hexadecimal digit // J8 is not a valid Hex number

One mark for each combination of valid or invalid **and** the reason.**(c) Two points from:****[2]**

- Public address can be reached across the Internet.
- Private address can only be reached internally/through the LAN/Intranet // private address cannot be reached across the Internet.
- NAT (Network Address Translation) is necessary for a private IP address to access the Internet directly.
- A private address is more secure than a public address // A public address is less secure than a private address.
- Public addresses are provided by ISP / assigned by InterNIC // Private addresses are assigned by the router (of the network concerned).
- Public addresses are unique (to the Internet) // Private addresses (are unique within their network, but) can be duplicated within other (discrete) networks.
- 10.0.0.1 to 10.255.255.254 and 172.16.0.1 to 172.31.255.254 and 192.168.0.1.to 192.168.255.254 form the private address space // IP addresses from the private address space are never assigned as public.

53. M/J 16/P12/Q7/a,b,c**7 (a) Internet Protocol****[1]**

(b)**[4]**

Address	Denary / Hexadecimal	Valid or Invalid	Reason
3.2A.6AA.BBBB	Hexadecimal	Invalid	One point from: - This is more than <u>32 bits</u> <u>6AA / BBBB</u> in Hex is bigger than <u>FF / 255</u> in denary <u>6AA / BBBB</u> uses more than 8 bits / a byte - The third / fourth group is bigger than <u>FF / 255</u> in denary - The third / fourth group uses more than 8 bits / a byte
2.0.255.1	Denary	Valid	There are 4 bytes, each 255 or below // All the values are in the range 0 - 255
6.0.257.6	Denary	Invalid	<u>257</u> is above 255 // The third group is above 255
0A.78.F4.J8	Hexadecimal	Invalid	J is not a valid hexadecimal digit // J8 is not a valid Hex number

One mark for each combination of valid or invalid **and** the reason.

(c) Two points from:**[2]**

- Public address can be reached across the Internet.
- Private address can only be reached internally/through the LAN/Intranet // private address cannot be reached across the Internet.
- NAT (Network Address Translation) is necessary for a private IP address to access the Internet directly.
- A private address is more secure than a public address // A public address is less secure than a private address.
- Public addresses are provided by ISP / assigned by InterNIC // Private addresses are assigned by the router (of the network concerned).
- Public addresses are unique (to the Internet) // Private addresses (are unique within their network, but) can be duplicated within other (discrete) networks.
- 10.0.0.1 to 10.255.255.254 and 172.16.0.1 to 172.31.255.254 and 192.168.0.1 to 192.168.255.254 form the private address space // IP addresses from the private address space are never assigned as public.

54. M/J 16/P13/Q6/a,b,c,d,e**6 (a) Two from:****[2]**

- WWW is a collection of interlinked, hypertext documents/webpages/multimedia resources (accessed via the Internet) // WWW is content from web servers organised as web pages
- Internet is the global connection of interconnected computer networks
- The Internet uses TCP/IP protocol / WWW uses http protocols to transmit data

(b)**[5]**

Description	Fibre-Optic cables	Copper cables	Radio waves
'Wireless' media			✓
Twisted-pair is an example		✓	
Uses light waves	✓		
WiFi			✓
Fastest transmission media	✓		

(c) One pair from:**[2]**

- Real-time - a live stream of an event that is currently taking place
- On-demand - streaming of an event/programme that has taken place in the past
- Real time – the event is captured live with a video camera connected to a computer
- On-demand – Existing media are encoded to bit streaming format and uploaded to a server
- Real-time – cannot be paused / rewind etc
- On-demand – can be paused / re-wound / fast forwarded etc

(d) Two marks for description, one mark for correct example.**[3]**

- Four numbers separated with '.'
- Each number is between 0 and 255 / 00 and FF in Hex / stored in one byte.
- 32 bits long
- Correct example

(e) Four from:**[4]**

- URL is a reference address to a resource on the Internet.
- The URL is passed to the nearest Domain Name Server (by browser software).
- DNS server stores a database / list of URLs and matching IP addresses.
- DNS (Name Resolver) looks for the URL in its database.
- Finds the matching IP address and returns it to the originator.
- Or if it cannot find it, it forwards to another Domain Name Server at a higher level.
- (Original) DNS server adds the returned IP address to its cache.
- (Original) DNS server returns the IP address to the browser.

55. O/N 15/P11/Q1/a,b,c**1 (a) any two from:**

- sequence of digital signals / bits
- over a communication path / Internet
- transfer of data at high speed
- requires fast broadband connection
- requires some form of buffering
- bits arrive in the same order as sent

[2]

(b) (i) any **two** from:

- no need to wait for a whole file to be downloaded
 - no need to store large files on user's computer
 - allows on demand playback
 - no specialist software is required for playback in browser
- [2]

(ii) any **two** from:

- video stops / hangs if very slow Internet / broadband speed low
 - video stops / hangs if inadequate buffering capacity
 - loss of Internet means can't access films / files
 - may require specific software to run the files / films
 - viruses can be downloaded from the websites
- [2]

(c) 2 marks for on-demand and 2 marks for real-time

on-demand

- digital video tape, analogue video tape, or digital files are converted to bit streaming – format for broadcasting on the net; this is known as encoding, these encoded streaming video files are then uploaded to a dedicated server
- a link for the encoded video is placed on a web site
- a user clicks on the link to download the encoded streaming video; the streamed video is then broadcast to the user as and when they require it
- can be paused / can go back and re-watch / fast-forward, etc.

real-time

- an event is captured live with a video camera
 - the video camera is connected to a computer
 - the video signal is converted to streaming media files (encoded) on the computer
 - the encoded feed is then uploaded from the computer to a dedicated streaming server via cable, DSL, or a high-speed internet connection
 - the server then sends the live images it to all users requesting it as real-time video streaming
 - cannot be paused etc.
- [4]

56. O/N 15/P11/Q3/a,b(i)

3 (a)

Statement	True (✓)
The IP address consists of any number of digits separated by single dots (.)	
Each number in an IP address can range from 0 to 255	✓
IP addresses are used to ensure that messages and data reach their correct destinations	✓
Public IP addresses are considered to be more secure than private IP addresses	

accept words TRUE or FALSE in right hand column

1 mark per tick, –1 mark for each wrong tick if more than 2

[2]

- (b) (i) http – enables browser to know what protocol is being used to access information in the domain

cie.org.uk – cie.org.uk is the domain name

computerscience.html – actual web page / file being viewed [3]

57. O/N 15/P11/Q6

- 6 any **four** points from (maximum 3 marks per type of cable):

- fibre optic cables have greater bandwidth
- fibre optic cables need less signal boosting // can transmit over longer distances
- fibre optic cables have greater security (more difficult to “tap” into)
- fibre optic cables are immune to electromagnetic and other effects
- fibre optic cabling is lighter in weight (easier to install)
- fibre optic cables consume less power
- copper cabling is less expensive to install
- copper cable is easier to install because it is more flexible
- it is easier to make terminations using copper cabling
- the expertise in use of copper cabling is more extensive
- has been around for years ... so very little is “unknown” about installations using this type of cabling [4]

58. O/N 15/P11/Q7/a,b

- 7 (a) (i) – at least one computer used to “serve” ...
– ... other computers are referred to as “clients”
– server provides services / applications etc. ...
– ... which may be requested by clients [2]

- (ii) any **two** from:

- files and resources are centralised
- creation of security / manage security
- user needs user name and password to access network
- centralised back-up
- intranet capability
- Internet monitoring
- clients can be less powerful machines, therefore less expensive to buy
- saving resources on server reduces the burden on the client [2]

- (b) router [1]

59. O/N 15/P13/Q1/a,b,c

- 1 (a) any **two** from:

- sequence of digital signals / bits
- over a communication path / Internet
- transfer of data at high speed
- requires fast broadband connection
- requires some form of buffering
- bits arrive in the same order as sent [2]

(b) (i) any **two** from:

- no need to wait for a whole file to be downloaded
 - no need to store large files on user's computer
 - allows on demand playback
 - no specialist software is required for playback in browser
- [2]

(ii) any **two** from:

- video stops / hangs if very slow Internet / broadband speed low
 - video stops / hangs if inadequate buffering capacity
 - loss of Internet means can't access films / files
 - may require specific software to run the files / films
 - viruses can be downloaded from the websites
- [2]

(c) 2 marks for on-demand and 2 marks for real-time

on-demand

- digital video tape, analogue video tape, or digital files are converted to bit streaming – format for broadcasting on the net; this is known as encoding these encoded streaming video files are then uploaded to a dedicated server
- a link for the encoded video is placed on a web site
- a user clicks on the link to download the encoded streaming video; the streamed video is then broadcast to the user as and when they require it
- can be paused / can go back and re-watch / fast-forward, etc.

real-time

- an event is captured live with a video camera
 - the video camera is connected to a computer
 - the video signal is converted to streaming media files (encoded) on the computer
 - the encoded feed is then uploaded from the computer to a dedicated streaming server via cable, DSL, or a high-speed internet connection
 - the server then sends the live images it to all users requesting it as real-time video streaming
 - cannot be paused etc.
- [4]

60. O/N 15/P13/Q3/a,b(i)

3 (a)

Statement	True (✓)
The IP address consists of any number of digits separated by single dots (.)	
Each number in an IP address can range from 0 to 255	✓
IP addresses are used to ensure that messages and data reach their correct destinations	✓
Public IP addresses are considered to be more secure than private IP addresses	

accept words TRUE or FALSE in right hand column

1 mark per tick, –1 mark for each wrong tick if more than 2

[2]

- (b) (i) http – enables browser to know what protocol is being used to access information in the domain

cie.org.uk – cie.org.uk is the domain name

computerscience.html – actual web page / file being viewed [3]

61. O/N 15/P13/Q6

6 any **four** points from (maximum 3 marks per type of cable):

- fibre optic cables have greater bandwidth
 - fibre optic cables need less signal boosting // can transmit over longer distances
 - fibre optic cables have greater security (more difficult to “tap” into)
 - fibre optic cables are immune to electromagnetic and other effects
 - fibre optic cabling is lighter in weight (easier to install)
 - fibre optic cables consume less power
 - copper cabling is less expensive to install
 - copper cable is easier to install because it is more flexible
 - it is easier to make terminations using copper cabling
 - the expertise in use of copper cabling is more extensive
 - has been around for years ... so very little is “unknown” about installations using this type of cabling
- [4]

62. O/N 15/P13/Q7/a,b

- 7** (a) (i) – at least one computer used to “serve” ...
– ... other computers are referred to as “clients”
– server provides services / applications etc. ...
– ... which may be requested by clients [2]

(ii) any **two** from:

- files and resources are centralised
 - creation of security / manage security
 - user needs user name and password to access network
 - centralised back-up
 - intranet capability
 - Internet monitoring
 - Clients can be less powerful machines, therefore less expensive to buy
 - Saving resources on server reduces the burden on the client
- [2]

(b) router [1]

63. M/J 15/P11/Q5/b,c(i,iii)

- (b) **maximum of two marks** for Internet references and **maximum of two marks** for world wide web references

Internet

- massive network of networks/interconnected network of computer devices
- Internet stands for Interconnected Networks
- uses TCP/IP protocol

World Wide Web (www)

- is a collection of (multimedia) web pages/documents
- ...stored on websites
- http/protocols used to transmit data
- web pages are written in HTML
- URLs specify the location of the web pages
- web documents are accessed using browsers

[3]

- (c) (i) router

[1]

- (iii) server

[1]**64. M/J 15/P12/Q5/b,c(i,iii)**

- (b) **maximum of two marks** for Internet references and **maximum of two marks** for world wide web references

Internet

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[3]

- (c) (i) router

[1]

- (iii) server

[1]